PARAHORIC-LEVEL FINITE-SLOPE P-ADIC L-FUNCTIONS FOR SHALIKA AUTOMORPHIC REPRESENTATIONS

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

Introduction

Two of the most stunning recent advances towards the Birch–Swinnerton-Dyer conjecture are Skinner's converse theorem [19] for certain arithmetic rank 1 elliptic curves and Loeffler–Zerbes' work [15] for certain analytic rank 0 abelian surfaces. In both cases, p-adic L-functions provide the crucial connection between the analytic and algebraic sides.

The present paper is part of a program aiming to attach p-adic L-functions to cuspidal automorphic representations π of GL_N over a totally real number field and study their properties, such as trivial zeros. We are focusing our study to the already large class of π which are regular algebraic (i.e. cohomological of weight λ) and essentially self-dual of symplectic type (i.e. admitting the so-called Shalika model) forcing $N = 2n$ to be even. In contrast to constructions based on the Rankin–Selberg method (e.g. Januszewski, Namikawa, Eischen–Harris–Li–Skinner for unitary groups, or Wan, Eischen–Wan, Rosso–Liu for symplectic groups) our approach does not require π_p to be ordinary, but only of finite slope.

In this paper, we provide the first construction of p-adic L-functions and p-adic families, for finite slope cuspidal automorphic representations of GL_N allowing ramification at p. More precisely we allow parahoric level both at p and in the tame level (with respect to the Siegel parabolic), which is the natural square-free level for studying p-adic L-functions of representations with Shalika models (see [4, §1.2.3] for a thorough discussion and numerical examples). To achieve this, we develop a theory of Shalika local coefficients analogous to Shahidi's classical theory of local coefficients [18], and prove a consequential factorization result that allows us to construct an improved Ash–Ginzburg Shalika functional for parahoric spherical local representations. In passing, we show existence of explicit Friedberg–Jacquet test vectors (previously only available in the spherical case) allowing us to computing the local zeta integrals at all finite order characters. Using the results of [4], this informs the geometry of the parabolic eigenvariety at new points and allows to construct a two-variable p-adic L-functions with controlled growth and satisfying the precise interpolation relations conjectured by Coates and Perrin-Riou (formulated originally in [8] for good reduction at p, but later observed by Hansen that the statement makes sense more generally).

Theorem 1. *Let π be a RASCAR on GL_{2n} over a totally real field F which is parahoric spherical at all finite places, and let $\widetilde{\pi} = (\pi, (\alpha_{\mathfrak{p}})_{\mathfrak{p}|p})$ be regular parahoric refinement which is non-critical. There exists an admissible distribution $\mathcal{L}_p(\widetilde{\pi})$ on Gal_p (the Galois group of the maximal abelian extension of F unramified outside $p\infty$) of controlled growth such that for every finite order Hecke character χ of Gal_p and all $j \in \mathrm{Crit}(\lambda)$*

$$\iota_p^{-1}(\mathcal{L}_p(\tilde{\pi}, \chi\chi_{\mathrm{cyc}}^j)) = \mathcal{G}(\chi_f)^n \mathrm{N}_{F/\mathbb{Q}}(-i\mathfrak{d})^{jn} e_p(\tilde{\pi}_p, \chi, j) \cdot \frac{L^{(p)}\big(\pi \otimes \chi, j + \frac{1}{2}\big)}{\Omega^{\epsilon}_{\tilde{\pi}}}, \tag{1}$$

where $\epsilon = (\chi\chi_{\mathrm{cyc}}^j \eta)_\infty$, $\mathcal{G}(\chi_f)$ is the Gauss sum and $e_p(\tilde{\pi}_p, \chi, j)$ is a local factor (see Theorem 19)

The use of the Friedberg–Jacquet integral representation of the standard L-function of π for p-adic interpolation of the critical values was initiated in Ash–Ginzburg [2] and further developed in [9]. The works [4, 3] provide a fairly complete construction of p-adic L-functions in families based on Stevens' theory of

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overconvergent cohomology and use them to prove profound results on the geometry of the GL_{2n} eigenvariety, including étalness over the pure weight space and density of classical points of Shalika type. Previous results, however, crucially relied on the existence of explicit Friedberg–Jacquet test vectors, limiting most of the results to π everywhere unramified.

In this paper, we succeed in constructing p-adic L-functions beyond level 1 by generalizing Shahidi's work on local coefficients, which arose in his construction of general automorphic L-function [18], to the setting of Shalika models. Shahidi's local coefficients appear as Whittaker model multiplication factors in the composition of simple intertwining operators, which provide the necessary Euler factors of automorphic L-functions. In the case of automorphic representations with Shalika models, our local coefficients generate the local Euler factors of the exterior square automorphic L-function. This factorization result, which is the main result of §2, corroborates the relationship between exterior square automorphic L-functions and Shalika models.

In appearance of technical nature, the improvements of Theorem 1 are crucial for the arithmetic applications we aim at, namely the Greenberg–Benois trivial zero conjecture at the central point for $L_p(\tilde{\pi}, s)$. Indeed, the occurrence of a central trivial zero forces π_p to ramify and the conjectural formula requires evaluations at the trivial character. Allowing parahoric tame level also significantly broadens the scope of application of our results, the previous construction of p-adic L-functions in families being unconditional only for level 1.

Remark 2. In order to obtain precise interpolation formulas à la Coates–Perrin-Riou, we must start with the Ash–Ginzburg Shalika functional on the full induced representation, which is only defined if the character induced is regular and ordered in a particular way. It is this requirement that forces us to restrict our attention to regular parahoric spherical representations: they are the only ones that can be realized as subrepresentations of such Ash–Ginzburg ordered characters.

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1. Parahoric spherical representations admitting a Shalika model

In this section, we specify the relationship between parahoric invariants and Shalika models for generic representations.

Let $F/\mathbb{Q}_p$ be a finite extension with ring of integers $\mathcal{O}$, uniformizer ϖ and different ideal (ϖ^δ). Let B be the upper triangular Borel subgroup of $G = \mathrm{GL}_{2n}(F)$. The Iwahori subgroup $I \subset \mathrm{GL}_{2n}(\mathcal{O})$ consists of matrices which are upper triangular mod ϖ. Let $M_n = M_n(F)$ and B_n the Borel subgroup of $G_n = \mathrm{GL}_n(F)$.

We fix an additive character $\psi : F \to \mathbb{C}^\times$ with kernel $(\varpi^{-\delta})$ and we recall that the Shalika subgroup $S = G_n \ltimes M_n$ of G is endowed with a character ψ sending $s = \left(\begin{smallmatrix} h & \\ & h \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & X \\ & 1 \end{smallmatrix}\right) \in S$ to $\psi(\mathrm{tr}(X))$. Note that as the object of our study is parahoric spherical representations, we can restrict our local study to the case $\eta = \mathbf{1}$.

We say that an an irreducible smooth representation π of G admits a Shalika model if can be embedded G-equivariently in $\mathrm{Ind}_S^G \psi$, or equivalently by Frobenius reciprocity, if there exists a non-zero map $\mathcal{S}_\pi : \pi \to \mathbb{C}$ such that $\mathcal{S}_\pi(s \cdot f) = \psi(s)\mathcal{S}_\pi(f)$ for all $s \in S$. Note that by the Multiplicity One Theorem for Shalika

models [12, Prop. 6.1], if such a functional exists then it is unique, up to a non-zero scalar. Given a Shalika functional $\mathcal{S}_\pi$, the Shalika function attached to $f \in \pi$ is

$$W_f : G \to \mathbb{C}, \quad g \mapsto \mathcal{S}_\pi(g \cdot f),$$

and $f \mapsto W_f$ is the embedding of π in $\mathrm{Ind}_S^G \psi$. The work of Friedberg–Jacquet [10] gives an integral representation for the L-function of π in terms of its Shalika model. Namely, for $W \in \mathrm{Ind}_S^G \psi$, the zeta integral

$$\zeta(\chi, W, s) = \int_{G_n} W(\begin{pmatrix} h & \\ & \mathbf{1} \end{pmatrix})\chi(\det h)|\det h|^{s-1/2} dh$$

converges absolutely for $\mathrm{Re}(s) \gg 0$ and there exists a test vector W_π such that $\zeta(W_\pi, \chi, s) = L(\pi \otimes \chi, s)$.

A result of Matringe [16, Cor. 1.1] describes all generic representations π admitting Shalika models. For instance, when π is Iwahori spherical and generic, π has a Shalika model if, and only if, π is isomorphic to the normalized parabolic induction

$$\mathrm{Ind}_P^G \left(\prod_i (\theta_i \times \theta_i^{-1}) \times \prod_{m=2}^{n} \prod_j (\eta_{j,m} \otimes \mathrm{St}_m \times \eta_{j,m}^{-1} \otimes \mathrm{St}_m) \times \prod_{m=1}^{n} \prod_j \varepsilon_{j,m} \otimes \mathrm{St}_{2m} \right) \tag{2}$$

where the $\eta_{j,m}$'s are arbitrary characters, while the $\varepsilon_{j,m}$'s are trivial or quadratic, and where St_m is the GL_m-Steinberg and P is an appropriate parabolic so that the induction makes sense. We will see in Proposition 6 that π is parahoric spherical if and only if the only Steinberg representations occurring in (2) are St_2's.

We now introduce a class of parabolic subgroups of Levi subgroups of G. Let Δ be the positive roots of G corresponding to B. If $T \subset S \subset \Delta$, we denote $P_{S,T}$ the subgroup of G with Lie algebra

$$\mathfrak{p}_{S,T} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \mathbb{Z}_+ S - \mathbb{Z}_+ T} \mathfrak{g}_\alpha.$$

We remark that $P_{\Delta,\varnothing} = B$, $P_{\Delta,\Delta} = G$, while $P_{\Delta,S}$ is a usual parabolic subgroup, with Levi equal to $P_{S,S}$. Let

$$J_{S,T} = \{g \in P_{S,S}(\mathcal{O}) \mid g \mod \varpi \in P_{S,T}(\mathcal{O}/\varpi)\},$$

which specializes to $J_{\Delta,\varnothing} = I$ and $J_{\Delta,\Delta} = K$. Let W denote the Weyl group of G.

For a subset $S \subset \Delta$, we denote W_S the Weyl group of $P_{S,S}$. For $S, T \subset \Delta$ by [7, Prop 1.3.1] we have

$$P_{\Delta,S} \backslash G / P_{\Delta,T} \cong W_S \backslash W / W_T,$$

the latter having representatives $\{w \in W \mid wT > 0, w^{-1}S > 0\}$. It follows from the Iwahori decomposition that

$$P_{\Delta,S} \backslash G / J_{\Delta,T} \cong W_S \backslash W / W_T.$$

Lemma 3. *Let $w \in W$, chosen such that $wT > 0$ and $w^{-1}S > 0$. Then*

$$\mathrm{Levi}(w^{-1} P_{\Delta,S} w) \cap J_{\Delta,T} = J_{w^{-1}S, T \cap w^{-1}S}.$$

Proof. First, $w^{-1} P_{\Delta,S} w = P_{w^{-1}\Delta, w^{-1}S}$. If $g \in w^{-1} P_{\Delta,S} w \cap J_{\Delta,T}$, then $(g \mod \varpi) \in P_{w^{-1}\Delta, w^{-1}S} \cap P_{\Delta,T}$. As

$$\mathfrak{p}_{w^{-1}\Delta, w^{-1}S} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \mathbb{Z}_+ w^{-1}\Delta - \mathbb{Z}_+ w^{-1}S} \mathfrak{g}_\alpha \quad \text{and} \quad \mathfrak{p}_{\Delta,T} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \mathbb{Z}_+ \Delta - \mathbb{Z}_+ T} \mathfrak{g}_\alpha$$

their intersection equals $\mathfrak{t} \oplus \bigoplus \mathfrak{g}_\alpha$, where the direct sum is taken over $\alpha \in (\mathbb{Z}_+ w^{-1}\Delta - \mathbb{Z}_+ w^{-1}S) \cap (\mathbb{Z}_+ \Delta - \mathbb{Z}_+ T)$.

Any α in this intersection can be written in an irreducible way as

$$\alpha = \alpha_+ - \alpha_T = w^{-1}\beta_+ - w^{-1}\beta_S,$$

where $\alpha_+, \beta_+ \in \mathbb{Z}_+\Delta$, $\alpha_T \in \mathbb{Z}_+ T$ and $\beta_S \in \mathbb{Z}_+ S$. As $w^{-1}S > 0$ we deduce $w^{-1}\beta_S \leqslant \alpha_T$. Applying w we also get

$$w\alpha_+ - w\alpha_T = \beta_+ - \beta_S,$$

and this time $wT > 0$ implies that $\alpha_T \leqslant w\beta_S$. We deduce that $\alpha_+ = w^{-1}\beta_+ \in \Delta \cap w^{-1}\Delta$ and $\alpha_T = w^{-1}\beta_S \in T \cap w^{-1}S$, and so

$$\mathfrak{p}_{w^{-1}\Delta, w^{-1}S} \cap \mathfrak{p}_{\Delta,T} = \mathfrak{p}_{\Delta \cap w^{-1}\Delta, T \cap w^{-1}S}.$$

Further restricting to the Levi subgroup implies the desired identity of groups. □

Lemma 4. *Assume π as in (2) with $P = P_{\Delta,S}$. If π has non-zero invariants by $J_{\Delta,T}$, for some T, then there exists $w \in W$ such that $wT > 0$, $w^{-1}S > 0$ and $T \cap w^{-1}S = \varnothing$.*

Proof. By Mackey's formula, Frobenius reciprocity and Lemma 3, we have

$$\pi^{J_{\Delta,T}} = \mathrm{Hom}_{J_{\Delta,T}}(\mathbf{1}, \mathrm{Ind}_{P_{\Delta,S}}^{G}\sigma) = \bigoplus_{[w]\in P_{\Delta,S}\backslash G/J_{\Delta,T}} \mathrm{Hom}_{J_{\Delta,T}}(\mathbf{1}, \mathrm{Ind}_{(w^{-1}P_{\Delta,S}w)\cap J_{\Delta,T}}^{J_{\Delta,T}}\sigma^{w})$$

$$= \bigoplus_{[w]\in W_S\backslash W/W_T} (\sigma^{w})^{\mathrm{Levi}(w^{-1}P_{\Delta,S}w)\cap J_{\Delta,T}} = \bigoplus_{[w]\in W_S\backslash W/W_T} (\sigma^{w})^{J_{w^{-1}S,T\cap w^{-1}S}} \cong \bigoplus_{[w]\in W_S\backslash W/W_T} \sigma^{J_{S,wT\cap S}}$$

It follows that $\sigma^{J_{S,wT\cap S}} \neq 0$ for some $w \in W$ such that $wT > 0$ and $w^{-1}S > 0$. As the Levi of $P_{\Delta,S}$ on which the representation $\sigma = \prod_i(\theta_i \otimes \theta_i^{-1}) \times \prod_j \sigma_j$ is defined is $P_{S,S} = \prod_i \mathrm{GL}(1) \times \mathrm{GL}(1) \times \prod_j \mathrm{GL}(m_j)$, we have

$$\sigma^{J_{S,wT\cap S}} = \sigma^{P_{S,S}\cap J_{S,wT\cap S}}.$$

The Iwahori $J_{S,\varnothing}$ being the largest subgroup of $P_{S,S}$ under which σ has non-zero invariants, it follows that $P_{S,S} \cap J_{S,wT\cap S} \subset J_{S,\varnothing}$. Hence $wT \cap S = \varnothing$ as desired. □

Corollary 5. *Assume π is as in (2). If $\pi^{J_{\Delta,T}} \neq 0$, where $P_{\Delta,T}$ has m Levi blocks. Then each special representation σ_i has Zelevinsky length $\leqslant m$, in other words it has monodromy rank $\leqslant m-1$.*

Proof. By a block of a subset of Δ we refer to the roots of a Levi block of the associated parabolic.

By Lemma 4, for some w we have $w^{-1}S \cap T = \varnothing$. Since w satisfies $wT > 0, w^{-1}S > 0$, this is equivalent to the condition that each block of T is disjoint from each block of $w^{-1}S$. Let $\mathcal{B}$ be the largest block of $w^{-1}S$, of dimension d, equal to the length of the largest Zelevinsky segment. We will verify that $d \leqslant m$.

Writing $e_1, \ldots, e_{2n}$ for the standard characters of T, the block $\mathcal{B}$ is generated by the positive roots

$$\begin{aligned}\alpha_1 &= e_{i_1} - e_{i_2}\\ \alpha_2 &= e_{i_2} - e_{i_3}\\ &\vdots\\ \alpha_{d-1} &= e_{i_{d-1}} - e_{i_d},\end{aligned}$$

in which case the positive roots of $\mathcal{B}$ are $\{e_{i_j} - e_{i_k} \mid j > k\}$. If $d > m$, by the pigeonhole principle there must exist some indices $j > k$ such that i_j and i_k lie in the same Levi block of T, in other words $e_{i_j} - e_{i_k}$ is also in a block of T, yielding a contradiction. □

As the Siegel parabolic subgroup has two Levi components Corollary 5 has the following consequence.

Proposition 6. *Let π be an generic smooth irreducible representation of G admitting a Shalika model. Then π is parahoric spherical if and only if each special representation appearing in the parabolic induction (2) is a twist of St_2 and any twist by a non-quadratic character appears accompanied by the inverse twist.*

Moreover $\dim \pi^J = 2^{n-\mathrm{rk}\, N_\pi}$. In particular, $\dim \pi^J = 1$ if and only if π is maximally Steinberg.

2. Improved Ash–Ginzburg functionals

Matringe's classification of generic representations of $G = \mathrm{GL}_{2n}(F)$ admitting Shalika models relies on an inductive construction of Shalika models which is hard to control for the computation of local zeta functions. In this section, we will explain how using intertwining operators one can rescale the Ash–Ginzburg integral yielding a Shalika functional on any parahoric spherical representations.

Haar measures. We fix a Haar measure dx on F giving $\mathcal{O}$ measure 1, which induces a measure $dX = \prod dx_{ij}$ on $M_n(F)$ giving $M_n(\mathcal{O})$ measure 1. The Haar measure dX satisfies $d(gX) = |\det g|^n dX$ and therefore yields the unimodular Haar measure $dX/|\det X|^n$ on $\mathrm{GL}_n(F)$. We denote by dg a rescaling of this measure giving $K_n = \mathrm{GL}_n(\mathcal{O})$ volume 1. The rescaling factor can be computed as the inverse of

$$\int_{K_n} dX/|\det X|^n = \int_{K_n} dX = \sum_{x\in \mathrm{GL}_n(\mathbb{F}_q)} \int_{x+\varpi M_n(\mathcal{O})} dX = q^{-n^2}\,|\mathrm{GL}_n(\mathbb{F}_q)| = \prod_{i=1}^{n}(1-q^{-i}).$$

The left Haar measure db on B_n is not unimodular and $d(hbh^{-1}) = \delta_B(h)db$, but its restriction to $B_n(\mathcal{O})$ is unimodular. We normalize db to give $B_n \cap K_n$ measure 1. In this case we obtain quotient measures dg on $B_n\backslash G_n$ and $B_n \cap K_n\backslash K_n$ which agree under the Cartan identification $B_n\backslash G_n = B_n \cap K_n\backslash K_n$.

2.1. **Statement of the local results.** Given a character $\theta = (\theta_1, \dots, \theta_{2n}) : T = (F^\times)^{2n} \to \mathbb{C}^\times$, the (unique) generic constituent π of the normalized parabolic induction $I(\theta) = \mathrm{Ind}_B^G(\theta)$ admits a Shalika model if, and only if, (up to a permutation of the indices) $\theta_i\theta_{n+i} = \mathbf{1}$ for all $1 \leqslant i \leqslant n$. Recall that by a result of Casselman, π is Iwahori spherical (i.e., admits non-zero fixed vectors by the Iwahori subgroup I of G) if, and only if, θ is unramified. Henceforth we assume θ to be unramified, regular (i.e., $\theta_i \neq \theta_j$ for all $1 \leqslant i < j \leqslant 2n$) and $\theta_i\theta_{n+i} = \mathbf{1}$ for all $1 \leqslant i < j \leqslant n$, later referred to as the Ash–Ginzburg ordering. These conditions on θ force (the unique) subrepresentation π of $I(\theta)$ to be parahoric spherical. Given such a character θ, using analytic continuation, Ash–Ginzburg [2] constructed a non-zero functional $\mathcal{S}^\theta_{\mathrm{AG}} : I(\theta) \to \mathbb{C}$ given by

$$\mathcal{S}^\theta_{\mathrm{AG}}(f) = \int_{G/B}\int_{M_n(F)} f\left(\begin{pmatrix} & \mathbf{1} \\ \mathbf{1} & X \end{pmatrix}\begin{pmatrix} h & \\ & h \end{pmatrix}\right)\psi(-\operatorname{tr} X)\,dX\,dh.$$

The Ash–Ginzburg functional has been instrumental in computing local zeta integrals for π spherical (see [2, 9, 4, 3]). If $I(\theta)$ is reducible with our generic π occurring as subrepresentation, then $\mathcal{S}^\theta_{\mathrm{AG}}|_\pi$ would still be a Shalika functional for π, provided it is non-zero. We determine that this is the case precisely when π has monodromy rank 1, allowing one to define $\mathcal{S}_\pi$ simply as $\mathcal{S}^\theta_{\mathrm{AG}}|_\pi$.

In the general case, we develop a theory of Shalika local coefficients and define $\mathcal{S}_\pi$ by analytic continuation of Ash–Ginzburg integrals rescaled by Shalika local coefficients. Note that, when the monodromy rank of π is $\geqslant 2$, then our methods imply the existence of some non-generic, non-tempered parahoric spherical representations which would be a theme interesting to pursue on its own.

Given a regular unramified character θ, for $\tau \in W$ we let $M_\tau : I(\theta^{\tau^{-1}}) \to I(\theta)$ be the intertwining operator.

As $B\backslash G/J \cong W/W_H$, a basis of $I(\theta)^J$ is given by $\{f^\theta_{BwJ}, [w] \in W/W_H\}$, where $f^\theta_{BwJ} : G \to \mathbb{C}$ is the unique function supported on BwJ, normalized so that $f^\theta_{BwJ}\left(\begin{pmatrix} 1 & \\ & \varpi^{-\delta} \end{pmatrix} w\right) = 1$. Our focus now is to determine explicit vectors in π^J in the above decomposition.

The proof of the following result will occupy the entire section and be completed in §2.6.

Theorem 7. *Let $\pi \subset \mathrm{Ind}_B^G\,\theta$ be a parahoric spherical representation, with θ unramified, regular and such that $\theta_i\theta_{n+i} = \mathbf{1}$ for all $1 \leqslant i \leqslant n$. For each U-eigenvalue α on π^J, there exists a Weyl element τ such that $\pi = \operatorname{Im} M_\tau$, and a Shalika functional $\mathcal{S}_\pi : \pi \to \mathbb{C}$ such that $\phi^\tau_\pi = M_\tau(f^{\theta^{\tau^{-1}}}_{Bw_0J}) \in \pi^J[U - \alpha]$ such that $\mathcal{S}_\pi(\phi^\tau_\pi)(t^{-\delta}) = 1$.*

As a consequence we show existence of explicit Friedberg–Jacquet test vectors beyond the spherical case.

Proposition 8. *Suppose $\widetilde{\pi} = (\pi, \alpha)$ and ϕ^τ_π are as in Theorem 7. Letting $W_{\widetilde{\pi}} = \mathcal{S}_\pi(\phi^\tau_\pi)$, one has*

(1) If χ is an unramified character then

$$\zeta(s, W_{\widetilde{\pi}}, \chi) = q^{\delta s n - \delta n^2/2}\left(1 - \frac{1}{q}\right)^n \chi(\varpi)^{-\delta n}\prod_{i=1}^n L(\theta^{\tau^{-1}}_{n+i}\chi, s).$$

(2) If χ is any character and β is the conductor of χ (if ramified, otherwise $\beta = 1$) then

$$\zeta(s, u^{-1}t^\beta \cdot W_{\widetilde{\pi}}, \chi) = e_p(\widetilde{\pi}, \chi, s).$$

Proof. The latter formula follows from [3, Prop. 9.3]. The former is similar and we include the computations for the convenience of the reader. Writing $\theta' = \theta^{\tau^{-1}}$, $F_0 = f^{\theta'}_{Bw_0J}$ and recalling that $W_{\widetilde{\pi}} = \mathcal{S}^{\theta'}_{\mathrm{AG}}(F_0)$, we have

$$\begin{aligned}
\zeta(s, W_{\widetilde{\pi}}, \chi) &= \int_{G_n} W_{\widetilde{\pi}}\left[\begin{pmatrix} h & \\ & 1 \end{pmatrix}\right]\chi(\det h)|\det h|^{s-1/2}dh \\
&= \int_{G_n}\int_{K_n}\int_{M_n} F_0\left[\begin{pmatrix} 1 & \frac{1}{X} \end{pmatrix}\begin{pmatrix} kh & \\ & k \end{pmatrix}\right]\chi(\det h)|\det h|^{s-1/2}\psi\big(-\operatorname{tr} X\big)\,dX\,dk\,dh \\
&= \int_{G_n}\int_{K_n}\int_{M_n} F_0\left[\begin{pmatrix} 1 & \frac{1}{X} \end{pmatrix}\begin{pmatrix} h & \\ & 1 \end{pmatrix}\right]\chi(\det h)|\det h|^{s-1/2}\psi\big(-\operatorname{tr} X\big)\,dX\,dk\,dh,
\end{aligned}$$

using that $\det khk^{-1} = \det h$, $d(khk^{-1}) = dh$, and the J-invariance of F_0. Writing $h = bk$ with $dh = dk \cdot db$ and using the fact that χ is unramified we get

$$\begin{aligned}
\zeta(s, W_{\widetilde{\pi}}, \chi) &= \int_{B_n}\int_{M_n} F_0\left[\begin{pmatrix} 1 & \frac{1}{X} \end{pmatrix}\begin{pmatrix} b & \\ & 1 \end{pmatrix}\right]\chi(\det b)|\det b|^{s-1/2}\psi\big(-\operatorname{tr} X\big)\,dX\,db \\
&= \int_{B_n}\int_{M_n} (\delta_B^{1/2}\theta')\left[\begin{pmatrix} 1 & \\ & b \end{pmatrix}\right]F_0\left[\begin{pmatrix} 1 & \\ & b^{-1}\frac{1}{X} \end{pmatrix}\right]\chi(\det b)|\det b|^{s-1/2}\psi\big(-\operatorname{tr} X\big)\,dX\,db,
\end{aligned}$$

Changing variables so that $b^{-1}X$ becomes X, noting that $d(bX) = |\det b|^n dX$, this equals

$$\zeta(s, W_{\widetilde{\pi}}, \chi) = \int_{B_n} \int_{M_n} \delta_B^{1/2} \theta'\left[\left(\begin{smallmatrix} 1 & \\ & b \end{smallmatrix}\right)\right] F_0\left[\left(\begin{smallmatrix} 1 & \frac{1}{X} \end{smallmatrix}\right)\right] \chi(\det b) |\det b|^{s+n-1/2} \psi\big(-\operatorname{tr} bX\big)\, dX\, db.$$

As in [3, Prop. 9.3], $X \in M_n(\mathcal{O})$ and thus $b \in B_n(F) \cap \varpi^{-\delta} M_n(\mathcal{O})$. Since $F_0\left(\left(\begin{smallmatrix} 1 & \\ & \varpi^{-\delta} \end{smallmatrix}\right) w_0\right) = 1$, we have $F_0(w_0) = \delta_B^{\frac{1}{2}} \theta'\left(\left(\begin{smallmatrix} 1 & \\ & \varpi^{\delta} \end{smallmatrix}\right)\right)$, so after integrating out X we get

$$\begin{aligned}\zeta(s, W_{\widetilde{\pi}}, \chi) &= \int_{B_n \cap \varpi^{-\delta} M_n(\mathcal{O})} \delta_B^{1/2} \theta'\left[\left(\begin{smallmatrix} 1 & \\ & \varpi^{\delta} b \end{smallmatrix}\right)\right] \chi(\det b) |\det b|^{s+n-1/2}\, db \\ &= \delta_B^{\frac{1}{2}} \theta'\left(\left(\begin{smallmatrix} 1 & \\ & \varpi^{\delta} \end{smallmatrix}\right)\right) \prod_{i=1}^{n} \int_{\varpi^{-\delta}\mathcal{O}} \theta'_{n+i} \chi |\cdot|^{s-1}(t_i) dt_i = q^{\delta s n - \delta n^2/2} \left(1 - \tfrac{1}{q}\right)^n \chi(\varpi)^{-\delta n} L(\theta'_{n+i}\chi, s).\end{aligned}$$

Here we used that if we write $b = \begin{pmatrix} t_1 & & u_{ij} \\ & \ddots & \\ 0 & & t_n \end{pmatrix}$, then $db = \prod |t_i|^{i-n-1} \prod dt_i \prod du_{ij}$. □

2.2. **Shalika local coefficients.** Suppose θ an unramified regular character with the Ash–Ginzburg ordering and $\tau \in W$ such that $\theta^{\tau^{-1}}$ is also in the Ash–Ginzburg ordering. If $I(\theta)$ is a irreducible then the intertwining operator $M_\tau : I(\theta^{\tau^{-1}}) \to I(\theta)$ is an isomorphism, yielding a diagram

$$\begin{array}{ccc} I(\theta^{\tau^{-1}}) & \xrightarrow{\quad M_\tau \quad} & I(\theta) \\ & {\scriptstyle \mathcal{S}^{\theta^{\tau^{-1}}}_{\mathrm{AG}}} \searrow \quad \swarrow {\scriptstyle \mathcal{S}^{\theta}_{\mathrm{AG}}} & \\ & \operatorname{Ind}_S^G \psi & \end{array}$$

By multiplicity one for Shalika models, there exists a constant $\mathcal{C}_{\theta,\tau}$ such that $\mathcal{S}_\theta \circ M_\tau = \mathcal{C}_{\theta,\tau} \mathcal{S}_{\theta^{\tau^{-1}}}$.

Whenever $\ell(\tau\tau') = \ell(\tau) + \ell(\tau')$, $M_{\tau'\tau} = M_\tau \circ M_{\tau'}$ and therefore $\mathcal{C}_{\theta,\tau'\tau} = \mathcal{C}_{\theta^{\tau^{-1}},\tau'} \mathcal{C}_{\theta,\tau}$. We conclude that, to determine $\mathcal{C}_{\theta,\tau}$, it suffices to compute it for τ of the following forms:

(1) $\tau = (i, i+1)(n+i, n+i+1)$ for some $i < n$, and
(2) $\tau = (n, 2n) = \nu^{-1}(n, n+1)\nu$, where $\nu = (n+1, n+2)\cdots(2n-1, 2n)$.

2.3. **The image of the open cell under intertwining operators.** For θ unramified regular character of T, we let

$$c_{i,j}(\theta) = \frac{1 - q^{-1}\theta_i\theta_j^{-1}(\varpi)}{1 - \theta_i\theta_j^{-1}(\varpi)} \quad \text{and} \quad d_{i,j}(\theta) = 1 - c_{j,i}(\theta) = c_{i,j}(\theta) - q^{-1} = \frac{1 - q^{-1}}{1 - \theta_i\theta_j^{-1}(\varpi)}.$$

We begin by computing the image of f^θ_{BwJ} under a simple intertwining operator.

Lemma 9. *Suppose $\alpha = (i, i+1) \in W$ is simple, and $w \in W/W_H$. If $w^{-1}\alpha w \in W_H$ then*

$$M_\alpha f^\theta_{BwJ} = c_{i,i+1}(\theta) f^{\alpha\theta}_{BwJ}.$$

Otherwise,

$$M_\alpha f^\theta_{BwJ} = \begin{cases} -d_{i+1,i}(\theta) f^{\alpha\theta}_{BwJ} + q^{-1} f^{\alpha\theta}_{B\alpha wJ} & \text{if } \ell(\alpha w) > \ell(w), \\ d_{i,i+1}(\theta) f^{\alpha\theta}_{BwJ} + f^{\alpha\theta}_{B\alpha wJ} & \text{if } \ell(\alpha w) < \ell(w). \end{cases}$$

Proof. By [6, Thm. 3.4], if α is simple and $w \in W$ then

$$M_\alpha f^\theta_{BwI} = \begin{cases} (c_{i,i+1}(\theta) - 1) f^{\alpha\theta}_{BwI} + q^{-1} f^{\alpha\theta}_{B\alpha wI} & \text{if } \ell(\alpha w) > \ell(w), \\ (c_{i,i+1}(\theta) - q^{-1}) f^{\alpha\theta}_{BwI} + f^{\alpha\theta}_{B\alpha wI} & \text{if } \ell(\alpha w) < \ell(w). \end{cases}$$

Therefore, the parahoric function $M_\alpha f^\theta_{BwJ}$ has support in $BwJ \cup B\alpha wJ$. In the latter case, the two cells are disjoint, while in the former they coincide. To compute the coefficients, it is enough to plug in w into the above formula. □

Proposition 10. *Let θ be an unramified regular character of T. For all $1 \leqslant i \leqslant n-1$, we have*

$$M_{(i,n)(n+i,2n)} f^{\theta}_{Bw_0J} = c_{i,n}(\theta)c_{n+i,2n}(\theta) \prod_{i<j<n} c_{i,j}(\theta)c_{n+i,n+j}(\theta)c_{j,n}(\theta)c_{n+j,2n}(\theta) f^{\theta^{(i,n)(n+i,2n)}}_{Bw_0J}.$$

Moreover, letting $\theta' = \theta^{(n,2n)} = (\theta_1, \cdots, \theta_{n-1}, \theta_{2n}, \theta_{n+1}, \cdots, \theta_{2n-1}, \theta_n)$, we have

$$\begin{aligned} M_{\nu^{-1}(n,n+1)\nu} f^{\theta}_{Bw_0J} &= \prod_{i=1}^{n-1} c_{n+i,2n}(\theta) \left(\prod_{i=1}^{n-1} c_{(n,n+i)}(\theta) d_{n,2n}(\theta) F_0^{\theta'} + \sum_{k=1}^{n-1} \prod_{i=k+1}^{n-1} c_{(n,n+i)}(\theta) d_{n,n+k}(\theta) F_k^{\theta'} + F_n^{\theta'} \right) \\ &= \prod_{i=1}^{n-1} c_{n+i,2n}(\theta) \left(F_n^{\theta'} + d_{n,2n-1}(\theta) F_{n-1}^{\theta'} + d_{n,2n-2}(\theta) c_{n,2n-1}(\theta) F_{n-2}^{\theta'} + \cdots \right), \end{aligned}$$

where for $1 \leqslant k \leqslant n$ we have denoted $F_k^{\theta} = f^{\theta}_{B(n+k,\ldots,n)w_0J} \in I(\theta)^J$.

Proof. The expression above follows inductively using the following three observations, the last two of which are direct consequences of Lemma 9:

- Every time we apply a simple transposition, the length decreases, as w_0 is the longest Weyl element,
- $M_\nu f^{\theta}_{Bw_0J} = \prod_{i=1}^{n-1} c_{n+i,2n}(\theta) f^{\nu\theta}_{Bw_0J}$, and
- for $0 \leqslant k \leqslant n$ we have $M_{(n+i,n+i+1)} F_k^{\theta} = c_{(n+i,n+i+1)}(\theta) F_k^{(n+i,n+i+1)\theta}$ if $i > k$ and

$$M_{(n+k,n+k+1)} F_k^{\theta} = (c_{(n+k,n+k+1)}(\theta) - q^{-1}) F_k^{(n+k,n+k+1)\theta} + F_{k+1}^{(n+k,n+k+1)\theta}.$$

Indeed, $(n, \ldots, n+k)(n+i, n+i+1)(n+k, \ldots, n) = (n+i, n+i+1) \in W_H$ for $i > k$, whereas $(n, \ldots, n+k)(n+k, n+k+1)(n_k, \ldots, n) = (n, n+k+1) \notin W_H$. □

2.4. **Shalika functions of certain test functions.** To determine Shalika local coefficients, it suffices to compute the value of Shalika functions as $t^{-\delta}\varpi^{\delta} = \left(\begin{smallmatrix} 1 & \\ & \varpi^{\delta} \end{smallmatrix}\right)$. Writing $\mathcal{D} = (\theta\delta_B^{1/2})^{w_0}(t^{-\delta}\varpi^{\delta}) = q^{-\delta n^2/2} \prod_{i=1}^{n} \theta_i(\varpi)^{\delta}$

$$\begin{aligned} \mathcal{S}_\theta(F_k^{\theta})(t^{-\delta}\varpi^{\delta}) &= \int_{M_n(F)} F_k^{\theta} \left(\begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} \begin{pmatrix} 1 & \\ & \varpi^{\delta} \end{pmatrix} \right) \psi(-\operatorname{tr} X) dX \\ &= \int_{M_n(F)} F_k^{\theta} \left(\begin{pmatrix} & 1 \\ 1 & \end{pmatrix} t^{-\delta}\varpi^{\delta} \begin{pmatrix} 1 & \varpi^{\delta} X \\ & 1 \end{pmatrix} \right) \psi(-\operatorname{tr} X) dX \\ &= q^{\delta n} \mathcal{D} \int_{M_n(F)} F_k^{\theta} \begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} \psi(-\operatorname{tr}(\varpi^{-\delta} X)) dX \end{aligned}$$

As F_k^{θ} is J-invariant, for $n \in N_n(\mathcal{O})$, we have $F_k^{\theta} \begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} = F_k^{\theta} \left(\begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} n^{-1} & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & X \\ & 1 \end{pmatrix} \begin{pmatrix} n & \\ & 1 \end{pmatrix} \right)$ yielding

$$\begin{aligned} \mathcal{S}(F_k^{\theta})(t^{-\delta}\varpi^{\delta}) &= q^{\delta n} \mathcal{D} \int_{M_n(F)} \int_{N_n(\mathcal{O})} F_k^{\theta} \left(\begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} n^{-1} & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & X \\ & 1 \end{pmatrix} \begin{pmatrix} n & \\ & 1 \end{pmatrix} \right) \psi(-\operatorname{tr}(\varpi^{-\delta} X)) dX dn \\ &= q^{\delta n} \mathcal{D} \int_{M_n(F)} \int_{N_n(\mathcal{O})} F_k^{\theta} \begin{pmatrix} & 1 \\ 1 & n^{-1} X \end{pmatrix} \psi(-\operatorname{tr}(\varpi^{-\delta} X)) dX dn \\ &= q^{\delta n} \mathcal{D} \int_{M_n(F)} F_k^{\theta} \left(\begin{smallmatrix} & 1 \\ 1 & X \end{smallmatrix} \right) \left(\int_{N_n(\mathcal{O})} \psi(-\operatorname{tr}(\varpi^{-\delta} n X)) dn \right) dX \end{aligned}$$

Observe that $\int_{N_n(\mathcal{O})} \psi(-\operatorname{tr}(\varpi^{-\delta} n X)) dn$ vanishes unless $X = (x_{ij})$ with $x_{ij} \in \mathcal{O}$ for all $i > j$. The J-invariance of F_k^{θ} allows to integrate these variables out, hence we may assume that X is upper triangular.

$$\mathcal{S}(F_k^{\theta})(t^{-\delta}\varpi^{\delta}) = q^{\delta n} \mathcal{D} \int_{B_n(F)} F_k^{\theta} \begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} \psi(-\operatorname{tr}(\varpi^{-\delta} X)) dX.$$

To compute the above integral, it will suffice to restrict out attention to those upper triangular X such that $\begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} \in \operatorname{supp}(F_k^{\theta}) = B(n+k, \ldots, n) w_0 J$. Suppose $\begin{pmatrix} a & b \\ & c \end{pmatrix} \in B$ such that

$$\begin{pmatrix} a & b \\ & c \end{pmatrix} \begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} = \begin{pmatrix} b & a+bX \\ c & cX \end{pmatrix} \in (n+k, \ldots, n) w_0 J = \begin{pmatrix} \varpi M_n(\mathcal{O}) & K_n \\ K_n & M_n(\mathcal{O}) \end{pmatrix}.$$

We deduce that

$$\left(\begin{array}{ccccccc|ccccccc}
\multicolumn{7}{c|}{b_{ij}, i<n} & \multicolumn{7}{c}{(a+bX)_{ij}, i<n} \\
\multicolumn{5}{c}{c_k} & \cdots & & \multicolumn{5}{c}{c_k x_k} & \cdots & \\
\hline
b_{n1} & b_{n2} & \cdots & & & & b_{nn} & \multicolumn{7}{c}{(a+bX)_{n,*}} \\
c_1 & c_{12} & \cdots & & & & c_{1n} & c_1x_1 & \cdots & & & & & \\
 & \ddots & & & & & & & \ddots & & & & & \\
 & & c_{k-1} & \cdots & & & c_{k-1,n} & & & c_{k-1}x_{k-1} & \cdots & & & \\
 & & & & c_{k+1} & \cdots & c_{k+1,n} & & & & & c_{k+1}x_{k+1} & \cdots & \\
 & & & & & \ddots & & & & & & & \ddots & \\
 & & & & & & c_n & & & & & & & c_nx_n
\end{array}\right)$$

is in $\begin{pmatrix} \varpi M_n(\mathcal{O}) & K_n \\ K_n & M_n(\mathcal{O}) \end{pmatrix}$. First, note that

$$\begin{pmatrix} c_{k+1} & \cdots & \\ & \ddots & \\ & & c_n \end{pmatrix} \in K_{n-k-1}, \quad \begin{pmatrix} c_{k+1}x_{k+1} & \cdots & \\ & \ddots & \\ & & c_nx_n \end{pmatrix} \in M_{n-k-1}(\mathcal{O})$$

which implies that $x_{ij} \in \mathcal{O}$ for $i > k$. As before, we can integrate these variables out and assume that $x_{ij} = 0$ for $i > k$. Similarly, we note that

$$\begin{pmatrix} b_{n1} & b_{n2} & \cdots & & b_{nn} \\ c_1 & c_{12} & \cdots & & c_{1n} \\ & \ddots & & & \\ & & c_{k-1} & \cdots & c_{k-1,n} \end{pmatrix} \in K_k, \quad \begin{pmatrix} & & & (a+bX)_{n,*} & \\ c_1x_1 & \cdots & & & \\ & \ddots & & & \\ & & c_{k-1}x_{k-1} & \cdots & \end{pmatrix} \in M_{k-1,n}(\mathcal{O}).$$

Dropping the last column of the matrix on the right means that we can ignore the upper triangular matrix a. We see that

$$\begin{pmatrix} & & & (bX)_{n,<n} \\ c_1x_1 & \cdots & & \\ & \ddots & & \\ & & c_{k-1}x_{k-1} & \cdots \end{pmatrix} = \begin{pmatrix} b_{n1} & b_{n2} & \cdots & b_{nk} \\ c_1 & c_{12} & \cdots & c_{1k} \\ & \ddots & & \\ & & c_{k-1} & c_{k-1,k} \end{pmatrix} \begin{pmatrix} x_1 & \cdots & & x_{1,n-1} \\ & \ddots & & \\ & & x_{k-1} & \cdots & x_{k-1,n-1} \end{pmatrix}$$

has integral entries.

Again, we deduce that $x_{ij} \in \mathcal{O}$ (and therefore can be assumed to be 0) for $i < k, j < n$.

Finally, from the parahoric matrix we deduce that the row $c_k, \ldots, c_{k,n}$ lies in (ϖ), and the row $0, \ldots, 0, c_k x_{k,n}$ is a row in K_n. Therefore, $c_k \in (\varpi)$ and $c_k x_{k,n} \in \mathcal{O}^\times$, which implies that $x_{k,n} \in F - \mathcal{O}$.

Putting everything together, we see that

$$\mathcal{S}(F_k^\theta)(t^{-\delta}\varpi^\delta) = q^{\delta n}\mathcal{D} \int_X F_k^\theta \left(\begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} \right) \psi(-\operatorname{tr}(\varpi^{-\delta}X))dX,$$

where the integral is taken over matrices X all of whose entries are 0 except the top k on the n-th column, the lowest entry of which is in $F - \mathcal{O}$.

What about this last column? We'll show that if X is of this form then $\begin{pmatrix} & 1 \\ 1 & X \end{pmatrix} \in B(n+k, \ldots, n)w_0J$ whenever $v(z_k) < 0$. In fact, we'll show that in this case we can choose our matrix $\begin{pmatrix} a & b \\ & c \end{pmatrix}$ such that

$$\begin{pmatrix} b_{n,1} & \cdots & & b_{n,k} \\ c_1 & \cdots & & c_{1,k} \\ & \ddots & & \\ & & c_{k-1} & c_{k-1,k} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{pmatrix} = \begin{pmatrix} -a_n \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

in which case, up to a unit, $z_k = c_1 \cdots c_{k-1}a_n$ and $c_k = z_k^{-1}$.

Suppose $v(z_k) < v(z_{k-1})$. Then we can choose $c_{k-1} = 1$, $c_{k-1,k} = -\frac{z_{k-1}}{z_k}$, in which case the above matrix identity becomes

$$\begin{pmatrix} b_{n,1} & \dots & b_{n,k-2} & b'_{n,k} \\ c_1 & \dots & c_{1,k-2} & c'_{1,k} \\ & \ddots & & \\ & & c_{k-2} & c'_{k-2,k} \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_{k-2} \\ z_k \end{pmatrix} = \begin{pmatrix} -a_n \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

where $b'_{n,k} = b_{n,k} + b_{n,k-1}\frac{z_{k-1}}{z_k}$ and $c'_{i,k} = c_{i,k} + c_{i,k-1}\frac{z_{k-1}}{z_k}$. The left matrix is still in K_{k-1} and (up to a unit) $z_k = ac_1 \cdots c_{k-2}$.

If $v(z_k) \geq v(z_{k-1})$, we can choose $c_{k-1} = -\frac{z_k}{z_{k-1}}$ and $c_{k-1,k} = 1$. Again, we get

$$\begin{pmatrix} b_{n,1} & \dots & b_{n,k-2} & b'_{n,k-1} \\ c_1 & \dots & c_{1,k-2} & c'_{1,k-1} \\ & \ddots & & \\ & & c_{k-2} & c'_{k-2,k-1} \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_{k-2} \\ z_{k-1} \end{pmatrix} = \begin{pmatrix} -a_n \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

where $b'_{n,k-1} = b_{n,k-1} + b_{n,k}\frac{z_k}{z_{k-1}}$ and $c'_{i,k-1} = c_{i,k-1} + c_{i,k}\frac{z_k}{z_{k-1}}$. The left matrix is still in K_{k-1} and (up to a unit) $z_{k-1} = ac_1 \cdots c_{k-2}$.

2.4.1. *The computation for F_k^θ when $k < n$.* In this case, $\operatorname{tr} X = 0$ and we are computing

$$\mathcal{S}_\theta(F_k^\theta)(t^{-\delta}\varpi^\delta) = q^{\delta n}\mathcal{D}\int_{v(z_k)<0} F_k^\theta \left(\begin{smallmatrix} 1 & \frac{1}{X} \end{smallmatrix}\right) dz_1 \dots dz_k. \tag{3}$$

For unramified characters $\eta, \eta_1, \dots, \eta_{k-1}$ let

$$\mathcal{I}(\eta, \eta_1, \dots, \eta_{k-1}) = \int_{v(z_k)<0} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\eta_i|\cdot|^{-(k-i)}(c_i)dz_1 \cdots dz_k,$$

in which case (3) becomes

$$\begin{aligned}
&= q^{\delta n}\int_{v(z_k)<0} \theta_n|\cdot|^{\frac{1}{2}}(a^{-1})\theta_{n+1}|\cdot|^{\frac{1}{2}-1}(c_1^{-1})\cdots\theta_{n+k-1}|\cdot|^{\frac{1}{2}-(k-1)}(c_{k-1}^{-1})\theta_{n+k}|\cdot|^{\frac{1}{2}-k}(ac_1\cdots c_k)dz_1\cdots dz_k \\
&= q^{\delta n}\int_{v(z_k)<0} \theta_n^{-1}\theta_{n+k}|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\theta_{n+i}^{-1}\theta_{n+k}|\cdot|^{i-k}(c_i)dz_1\cdots dz_k \\
&= q^{\delta n}\mathcal{I}(\frac{\theta_{n+k}}{\theta_n}, \frac{\theta_{n+k}}{\theta_{n+1}}, \dots, \frac{\theta_{n+k}}{\theta_{n+k-1}}).
\end{aligned}$$

Lemma 11. *We have $\mathcal{I}(\eta) = \left(1 - \frac{1}{q}\right)\frac{\eta(\varpi)^{-1}}{1-\eta(\varpi)^{-1}}$ and*

$$\mathcal{I}(\eta, \eta_1, \dots, \eta_{k-1}) = c(\eta_{k-1})\mathcal{I}(\eta, \eta_1, \dots, \eta_{k-2}) - d(\eta_{k-1})\mathcal{I}(\eta\eta_{k-1}^{-1}, \eta_1\eta_{k-1}^{-1}, \dots, \eta_{k-2}\eta_{k-1}^{-1}).$$

Proof. The first part is a straightforward computation. For the second part, we compute

$$\begin{aligned}
\mathcal{I}(\eta, \eta_1, \dots, \eta_{k-1}) &= \int_{v(z_k)<0, v(z_k)<v(z_{k-1})} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\eta_i|\cdot|^{i-k}(c_i)dz_1\cdots dz_k \\
&+ \int_{v(z_{k-1})\leqslant v(z_k)<0} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\eta_i|\cdot|^{i-k}(c_i)dz_1\cdots dz_k \\
&= \int_{v(z_k)<0, v(z_k)<v(z_{k-1})} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)dz_1\cdots dz_k \\
&+ \int_{v(z_{k-1})\leqslant v(z_k)<0} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)\eta_{k-1}|\cdot|^{-1}(-\frac{z_k}{z_{k-1}})dz_1\cdots dz_k \\
&= \int_{v(z_k)<0} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)dz_1\cdots dz_{k-2}dz_k\int_{v(z_k)<v(z_{k-1})} dz_{k-1} \\
&+ \int_{v(z_{k-1})<0} \eta|\cdot|^{-(k-1)}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k-1}(c_i)\eta_{k-1}(z_{k-1})^{-1}dz_1\cdots dz_{k-1}\int_{v(z_{k-1})\leqslant v(z_k)<0}\eta_{k-1}|\cdot|^{-1}(z_k)dz_k
\end{aligned}$$

$$= \int_{v(z_k)<0} \eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)|z_{k-1}\varpi|dz_1\cdots dz_{k-2}dz_k$$
$$+ \frac{1-q^{-1}}{1-\eta_{k-1}(\varpi)}\int_{v(z_{k-1})<0}\eta|\cdot|^{-(k-1)}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k-1}(c_i)(1-\eta_{k-1}(z_{k-1})^{-1})dz_1\cdots dz_{k-1}$$
$$= q^{-1}\mathcal{I}(\eta,\eta_1,\ldots,\eta_{k-2}) + \frac{1-q^{-1}}{1-\eta_{k-1}(\varpi)}\left(\mathcal{I}(\eta,\eta_1,\ldots,\eta_{k-2}) - \mathcal{I}(\eta\eta_{k-1}^{-1},\eta_1\eta_{k-1}^{-1},\ldots,\eta_{k-2}\eta_{k-1}^{-1})\right)$$
$$= \frac{1-q^{-1}\eta_{k-1}(\varpi)}{1-\eta_{k-1}(\varpi)}\mathcal{I}(\eta,\eta_1,\ldots,\eta_{k-2}) - \frac{1-q^{-1}}{1-\eta_{k-1}(\varpi)}\mathcal{I}(\eta\eta_{k-1}^{-1},\eta_1\eta_{k-1}^{-1},\ldots,\eta_{k-2}\eta_{k-1}^{-1}). \qquad \square$$

We conclude the following recursive formula:

$$q^{-\delta n}\mathcal{S}_\theta(F_k^\theta)(t^{-\delta}\varpi^\delta) = \mathcal{I}(\frac{\theta_{n+k}}{\theta_n},\frac{\theta_{n+k}}{\theta_{n+1}},\ldots,\frac{\theta_{n+k}}{\theta_{n+k-1}})$$
$$= c_{(n+k,n+k-1)}(\theta)\mathcal{I}(\frac{\theta_{n+k}}{\theta_n},\frac{\theta_{n+k}}{\theta_{n+1}},\ldots,\frac{\theta_{n+k}}{\theta_{n+k-2}}) - d_{(n+k,n+k-1)}(\theta)\mathcal{I}(\frac{\theta_{n+k-1}}{\theta_n},\frac{\theta_{n+k-1}}{\theta_{n+1}},\ldots,\frac{\theta_{n+k-1}}{\theta_{n+k-2}}).$$

2.4.2. *The computation for F_n^θ.* In this case, $\operatorname{tr} X = z_n$ and we are computing

$$\mathcal{S}_\theta(F_n^\theta)(t^{-\delta}\varpi^\delta) = q^{\delta n}\mathcal{D}\int_{v(z_n)<0} F_n^\theta\left(\begin{pmatrix}1 & \\ & X\end{pmatrix}\right)dz_1\ldots dz_n. \tag{4}$$

For unramified characters $\eta,\eta_1,\ldots,\eta_{k-1}$ let

$$\mathcal{I}_\psi(\eta,\eta_1,\ldots,\eta_{k-1}) = \int_{v(z_k)<0}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\eta_i|\cdot|^{i-k}(c_i)\psi(-z_k)dz_1\cdots dz_k.$$

In this case, we get

$$\mathcal{S}_\theta(F_n^\theta)(t^{-\delta}\varpi^\delta) = q^{\delta n}\int_{v(z_n)<0}\theta_n^{-1}\theta_{2n}|\cdot|^{-n}(a)\prod_{i=1}^{n-1}\theta_{n+i}^{-1}\theta_{2n}|\cdot|^{i-k}(c_i)\psi(-z_n)dz_1\cdots dz_k$$
$$= q^{\delta n}\mathcal{I}_\psi\left(\frac{\theta_{2n}}{\theta_n},\frac{\theta_{2n}}{\theta_{n+1}},\ldots,\frac{\theta_{2n}}{\theta_{2n-1}}\right).$$

As in the case $k<n$, we will compute this integral recursively:

Lemma 12. *We have $\mathcal{I}_\psi(\eta) = -q^{-1}\eta(\varpi)^{-1}$ and*

$$\mathcal{I}_\psi(\eta,\eta_1,\ldots,\eta_{k-1}) = q^{-1}\mathcal{I}_\psi(\eta,\eta_1,\ldots,\eta_{k-2}) - \frac{1}{q\eta_{k-1}(\varpi)}\mathcal{I}(\eta\eta_{k-1}^{-1},\eta_1\eta_{k-1}^{-1},\ldots,\eta_{k-2}\eta_{k-1}^{-1}).$$

Proof. The first part follows from the fact that (with $\delta=0$)

$$\int_{v(z)=m}\eta(z)\psi(-z)dz = \begin{cases}-\eta(\varpi)^{-1} & m=-1,\\ 0 & m<-1.\end{cases}$$

For the second part, we compute

$$\mathcal{I}_\psi(\eta,\eta_1,\ldots,\eta_{k-1}) = \int_{v(z_k)<0,v(z_k)<v(z_{k-1})}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\eta_i|\cdot|^{i-k}(c_i)\psi(-z_k)dz_1\cdots dz_k$$
$$+\int_{v(z_{k-1})\leqslant v(z_k)<0}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-1}\eta_i|\cdot|^{i-k}(c_i)\psi(-z_k)dz_1\cdots dz_k$$
$$= \int_{v(z_k)<0,v(z_k)<v(z_{k-1})}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)\psi(-z_k)dz_1\cdots dz_k$$
$$+\int_{v(z_{k-1})\leqslant v(z_k)<0}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)\eta_{k-1}|\cdot|^{-1}(-\frac{z_k}{z_{k-1}})\psi(-z_k)dz_1\cdots dz_k$$
$$= \int_{v(z_k)<0}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)\psi(-z_k)dz_1\cdots dz_{k-2}dz_k\int_{v(z_k)<v(z_{k-1})}dz_{k-1}$$

$$+\int_{v(z_{k-1})<0}\eta\eta_{k-1}^{-1}|\cdot|^{-(k-1)}(a)\prod_{i=1}^{k-2}\eta_i\eta_{k-1}^{-1}|\cdot|^{i-k-1}(c_i)dz_1\cdots dz_{k-1}\int_{v(z_{k-1})\leqslant v(z_k)<0}\eta_{k-1}|\cdot|^{-1}(z_k)\psi(-z_k)dz_k$$

$$=\int_{v(z_k)<0}\eta|\cdot|^{-k}(a)\prod_{i=1}^{k-2}\eta_i|\cdot|^{i-k}(c_i)|z_{k-1}\varpi|\psi(-z_k)dz_1\cdots dz_{k-2}dz_k$$

$$-\frac{1}{q\eta_{k-1}(\varpi)}\int_{v(z_{k-1})<0}\eta(a)\prod_{i=1}^{k-2}\eta\eta_{k-1}^{-1}|\cdot|^{-(k-1)}(a)\prod_{i=1}^{k-2}\eta_i\eta_{k-1}^{-1}|\cdot|^{i-k-1}(c_i)dz_1\cdots dz_{k-1}$$

$$=q^{-1}\mathcal{I}_\psi(\eta,\eta_1,\dots,\eta_{k-2})-\frac{1}{q\eta_{k-1}(\varpi)}\mathcal{I}(\eta\eta_{k-1}^{-1},\eta_1\eta_{k-1}^{-1},\dots,\eta_{k-2}\eta_{k-1}^{-1}).$$ □

Therefore

$$\begin{aligned}q^{-\delta n}\mathcal{S}_\theta(F_n^\theta)(t^{-\delta}\varpi^\delta)&=\mathcal{I}_\psi(\frac{\theta_{2n}}{\theta_n},\frac{\theta_{2n}}{\theta_{n+1}},\dots,\frac{\theta_{2n}}{\theta_{2n-1}})\\&=q^{-1}\mathcal{I}_\psi(\frac{\theta_{2n}}{\theta_n},\frac{\theta_{2n}}{\theta_{n+1}},\dots,\frac{\theta_{2n}}{\theta_{2n-2}})-\frac{\theta_{2n-1}}{\theta_{2n}}(\varpi)\mathcal{I}(\frac{\theta_{2n-1}}{\theta_n},\frac{\theta_{2n-1}}{q\theta_{n+1}},\dots,\frac{\theta_{2n-1}}{\theta_{2n-2}})\end{aligned}$$

2.5. **Factorization.** The main result of this section is a factorization of the Shalika local coefficient. We begin with a technical result on factoring rational functions subject to the recursion relations from §2.4.

Consider the rational functions $c(x)=\frac{1-q^{-1}x}{1-x}$, $d(x)=\frac{1-q^{-1}}{1-x}$ and $A(x_1,\dots,x_n)$ and $B(x_1,\dots,x_n)$ defined recursively by

$$\begin{aligned}A(x_1,\dots,x_n)&=c(x_n)A(x_1,\dots,x_{n-1})-d(x_n)A(x_1x_n^{-1},\dots,x_{n-1}x_n^{-1}) & A(x_1)&=-d(x_1)\\B(x_1,\dots,x_n)&=\frac{1}{q}B(x_1,\dots,x_{n-1})-\frac{1}{qx_n}B(x_1x_n^{-1},\dots,x_{n-1}x_n^{-1}) & B(x_1)&=-\frac{1}{qx_1}\end{aligned}$$

for $n\geq 2$, with the convention that $A(\varnothing)=1$.

Lemma 13. *For $n\geq 1$ we have*

$$B(x_1,\dots,x_n)-d\left(\frac{1}{qz}\right)^{-1}A(x_1z^{-1},\dots,x_nz^{-1})=-\frac{1}{x_1}c\left(\frac{1}{x_1}\right)c\left(\frac{x_2}{x_1}\right)\cdots c\left(\frac{x_n}{x_1}\right)\frac{1-x_1^{-1}}{z^{-1}-x_1^{-1}},\tag{5}$$

$$B(x_1,\dots,x_n)+\sum_{k=1}^{n}\left(d(x_k)\prod_{j=k+1}^{n}c(x_j)\right)A(x_1x_k^{-1},\dots,x_{k-1}x_k^{-1})=-\frac{1}{x_1}c\left(\frac{1}{x_1}\right)c\left(\frac{x_2}{x_1}\right)\cdots c\left(\frac{x_n}{x_1}\right).\tag{6}$$

Proof. The proof is a straightforward, but unelightening, induction. We include the computations for the convenience of the reader. We first prove (5) The base case $n=1$ is easy to check:

$$B(x_1)-d\left(\frac{1}{qz}\right)^{-1}A(x_1z^{-1})=-\frac{1}{qx_1}+\frac{1-\frac{1}{qz}}{1-\frac{1}{q}}\frac{1-\frac{1}{q}}{1-\frac{x_1}{z}}=-\frac{1}{qx_1}+\frac{1-\frac{1}{qz}}{1-\frac{x_1}{z}}=\frac{1-\frac{1}{qx_1}}{1-\frac{x_1}{z}}=-\frac{1}{x_1}c\left(\frac{1}{x_1}\right)\frac{1-x_1^{-1}}{z^{-1}-x_1^{-1}}.$$

The recursion relations give

$$\begin{aligned}B(x_1,\dots,x_{n+1})-d\left(\frac{1}{qz}\right)^{-1}A(x_1z^{-1},\dots,x_{n+1}z^{-1})&=\frac{1}{q}B(x_1,\dots,x_n)-\frac{1}{qx_{n+1}}B(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1})\\&\quad-d\left(\frac{1}{qz}\right)^{-1}c(x_{n+1}z^{-1})A(x_1z^{-1},\dots,x_nz^{-1})\\&\quad-d\left(\frac{1}{qz}\right)^{-1}d(x_{n+1}z^{-1})A(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1}).\end{aligned}$$

Using the inductive hypothesis, we have

$$A(x_1z^{-1},\dots,x_nz^{-1})=d(q^{-1}z^{-1})B(x_1,\dots,x_n)+\frac{d(q^{-1}z^{-1})}{x_1}c(x_1^{-1})c(x_2x_1^{-1})\dots c(x_nx_1^{-1})\frac{1-x_1^{-1}}{z^{-1}-x_1^{-1}}$$

$$A(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1})=d(q^{-1}x_{n+1}^{-1})B(x_1,\dots,x_n)+\frac{d(q^{-1}x_{n+1}^{-1})}{x_1}c(x_1^{-1})c(x_2x_1^{-1})\dots c(x_nx_1^{-1})\frac{1-x_1^{-1}}{x_{n+1}^{-1}-x_1^{-1}}$$

Plugging into the expression above gives

$$= B(x_1,\dots,x_n)\left(\frac{1}{q} + d(q^{-1}x_{n+1}^{-1})\left(-\frac{1}{qx_{n+1}} + \frac{d(x_{n+1}z^{-1})}{d(q^{-1}z^{-1})}\right) - c(x_{n+1}z^{-1})\right)$$
$$+ \frac{1}{x_1}c(x_1^{-1})c(x_2x_1^{-1})\cdots c(x_nx_1^{-1})\left(d(q^{-1}x_{n+1}^{-1})\left(-\frac{1}{qx_{n+1}} + \frac{d(x_{n+1}z^{-1})}{d(q^{-1}z^{-1})}\right)\frac{1-x_1^{-1}}{x_{n+1}^{-1}-x_1^{-1}} - c(x_{n+1}z^{-1})\frac{1-x_1^{-1}}{z^{-1}-x_1^{-1}}\right)$$

It's immediate to compute that the coefficient of $B(x_1,\dots,x_n)$ is

$$\frac{1}{q} + d(q^{-1}x_{n+1}^{-1})\left(-\frac{1}{qx_{n+1}} + \frac{d(x_{n+1}z^{-1})}{d(q^{-1}z^{-1})}\right) - c(x_{n+1}z^{-1}) = 0$$

and that

$$d(q^{-1}x_{n+1}^{-1})\left(-\frac{1}{qx_{n+1}} + \frac{d(x_{n+1}z^{-1})}{d(q^{-1}z^{-1})}\right)\frac{1-x_1^{-1}}{x_{n+1}^{-1}-x_1^{-1}} - c(x_{n+1}z^{-1})\frac{1-x_1^{-1}}{z^{-1}-x_1^{-1}} = -c(x_{n+1}z^{-1})\frac{1-x_1^{-1}}{z^{-1}-x_1^{-1}}$$

which finishes the inductive step.

We now turn to (6). Again, we proceed by induction, the base case being

$$B(x_1) + d(x_1) = -\frac{1}{x_1}c(x_1^{-1}).$$

By the inductive hypothesis, it suffices to verify that

$$B(x_1,\dots,x_{n+1}) + \sum_{k=1}^{n+1} d(x_k)\prod_{j=k+1}^{n+1} c(x_j)A(x_1x_k^{-1},\dots,x_{k-1}x_k^{-1}) =$$

$$= c(x_{n+1}x_1^{-1})\left(B(x_1,\dots,x_n) + \sum_{k=1}^{n} d(x_k)\prod_{j=k+1}^{n} c(x_j)A(x_1x_k^{-1},\dots,x_{k-1}x_k^{-1})\right)$$

By definition, it suffices to check

$$\frac{1}{q}B(x_1,\dots,x_n) - \frac{1}{qx_{n+1}}A(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1}) + \sum_{k=1}^{n+1} d(x_k)\prod_{j=k+1}^{n+1} c(x_j)A(x_1x_k^{-1},\dots,x_{k-1}x_k^{-1}) =$$

$$= c(x_{n+1}x_1^{-1})B(x_1,\dots,x_n) + c(x_{n+1}x_1^{-1})\sum_{k=1}^{n} d(x_k)\prod_{j=k+1}^{n} c(x_j)A(x_1x_k^{-1},\dots,x_{k-1}x_k^{-1})$$

which is equivalent to

$$\left(\frac{1}{q} - c(x_{n+1}x_1^{-1})\right)B(x_1,\dots,x_n) + \left(-\frac{1}{qx_{n+1}} + d(x_{n+1})\right)A(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1}) =$$

$$= \left(c(x_{n+1}x_1^{-1}) - c(x_{n+1})\right)\sum_{k=1}^{n-1} d(x_k)\prod_{j=k+1}^{n-1} A(x_1x_k^{-1},\dots,x_{k-1}x_k^{-1}).$$

A quick computation and another application of the inductive hypothesis leads to verifying

$$-d(x_{n+1}x_1^{-1})B(x_1,\dots,x_n) - \frac{1}{x_{n+1}}c(x_{n+1}^{-1})A(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1}) =$$

$$= \left(c(x_{n+1}x_1^{-1}) - c(x_{n+1})\right)\left(-\frac{1}{x_1}c(x_1^{-1})c(x_2x_1^{-1})\cdots c(x_nx_1^{-1}) - B(x_1,\dots,x_n)\right)$$

which is equivalent to

$$d(x_{n+1})B(x_1,\dots,x_n) + \frac{1}{x_{n+1}}c(x_{n+1}^{-1})A(x_1x_{n+1}^{-1},\dots,x_nx_{n+1}^{-1}) = \frac{1}{x_1}c(x_1^{-1})c(x_2x_1^{-1})\cdots c(x_nx_1^{-1})\frac{d(x_{n+1}^{-1})d(x_{n+1}x_1^{-1})}{d(x_1^{-1})},$$

which follows from the first part. □

We are now in a position to state the main factorization result:

Proposition 14. *We have*

$$\mathcal{C}_{\theta,(i,n)(n+i,2n)} = c_{n,i}(\theta)c_{2n,n+i}(\theta)\prod_{i<j<n} c_{n,j}(\theta)c_{2n,n+j}(\theta)c_{j,i}(\theta)c_{n+j,n+i}(\theta)$$

$$\mathcal{C}_{\theta,(n,2n)} = -\frac{\theta_n(\varpi)}{\theta_{2n}(\varpi)}c_{n,2n}(\theta)\prod_{i=1}^{n-1} c_{n,n+i}(\theta)c_{n+i,n}(\theta).$$

Proof. By definition of the Shalika local coefficient,

$$\mathcal{C}_{\theta,w} = \mathcal{C}_{\theta,w}q^{-\delta n}\mathcal{S}_{\theta^{w^{-1}}}f^{\theta^{w^{-1}}}_{Bw_0J}(t^\delta) = q^{-\delta n}\mathcal{S}_\theta M_{(n,2n)}f^{\theta^{w^{-1}}}_{Bw_0J}(t^{-\delta}\varpi^\delta)$$

The first part of the proposition follows immediately from Proposition 10. For the second part, note that

$$\mathcal{C}_{\theta,(n,2n)} = q^{-\delta n}\mathcal{S}_\theta M_{(n,2n)}f^{\theta^{(n,2n)}}_{Bw_0J}(t^{-\delta}\varpi^\delta)$$

By Proposition 10, this equals

$$\mathcal{C}_{\theta,(n,2n)} = \prod_{i=1}^{n-1} c_{n+i,n}(\theta)\left(q^{-\delta n}\mathcal{S}_\theta(F_n)(t^{-\delta}\varpi^\delta) + \sum_{k=0}^{n-1} d_{2n,n+k}\prod_{i=k+1}^{n-1} c_{2n,n+i}(\theta)q^{-\delta n}\mathcal{S}_\theta(F_k)(t^{-\delta}\varpi^\delta)\right)$$

The computations of §2.4.1 and §2.4.2 give

$$q^{-\delta n}\mathcal{S}_\theta(F_n)(t^{-\delta}\varpi^\delta) + \sum_{k=0}^{n-1} d_{2n,n+k}\prod_{i=k+1}^{n-1} c_{2n,n+i}(\theta)q^{-\delta n}\mathcal{S}_\theta(F_k)(t^{-\delta}\varpi^\delta)$$

$$= \mathcal{I}_\psi(\frac{\theta_{2n}}{\theta_n},\dots,\frac{\theta_{2n}}{\theta_{2n-1}}) + \sum_{k=0}^{n-1} d(\frac{\theta_{2n}}{\theta_{n+k}})\prod_{i=k+1}^{n-1} c(\frac{\theta_{2n}}{\theta_{n+i}})\mathcal{I}(\frac{\theta_{n+k}}{\theta_n},\dots,\frac{\theta_{n+k}}{\theta_{n+k-1}})$$

We now apply Lemma 13 with $x_i = \theta_{2n}\theta^{-1}_{n+i-1}(\varpi)$ and obtain that the above expression equals

$$-\frac{\theta_n(\varpi)}{\theta_{2n}(\varpi)}\prod_{i=1}^{n} c(\theta_n\theta^{-1}_{n+i}(\varpi)),$$

and the desired result follows. □

Remark 15. One might ask about the factorization of the Shalika local coefficient $\mathcal{C}_{\theta,(i,n+i)}$, which we have not computed. The complexity of the formulas involved are much greater. For instance, while $M_{(n,2n)}f^{\theta^{(n,2n)}}_{Bw_0J}$ is supported on $n+1$ parahoric cells, as in the proof above, we have

$$M_{(i,n+i)}f^{\theta^{(i,n+i)}}_{Bw_0J} = \prod_{u=1}^{i-1} c_{n+u,i}(\theta)\prod_{v=i+1}^{n} c_{n+i,v}(\theta)\left(d_{n+i,i}(\theta)\prod_{u=1}^{i-1} c_{n+i,n+u}(\theta)\prod_{v=i+1}^{n} c_{i,v}(\theta)f^\theta_{Bw_0J}\right.$$
$$\left. + \sum_{1\leqslant k\leqslant i\leqslant \ell\leqslant n} d_{\ell,i}d_{n+i,n+k}\prod_{u=i+1}^{\ell-1} c_{u,i}(\theta)\prod_{v=k+1}^{i-1} c_{n+i,n+v}(\theta)f^\theta_{B(\ell,\ell+1,\dots,k-1,k)w_0J}\right),$$

with the convention $d_{k,k}(\theta) = 1$. This function is supported on $i(n+1-i)+1$ cells, and the computation becomes daunting.

Remark 16. The coefficient $\mathcal{C}_{\theta,(n,2n)}$ can be written as

$$\mathcal{C}_{\theta,(n,2n)} = \gamma(\theta_n/\theta_{2n},1)\prod_{i=1}^{n-1}\gamma(\theta_n/\theta_{n+i},1)\gamma(\theta_{n+i}/\theta_n,1),$$

realizing the Shalika local coefficient as a subproduct of $\gamma(\wedge^2\pi,1)$.

2.6. **Proof of Theorem 7.** We are now ready to construct an explicit Shalika functional on parahoric spherical representations by analytic continuation. The idea is the following: restrict the Ash–Ginzburg Shalika integral to $\pi \subset \mathrm{Ind}_B^G\theta$. If the restriction does not vanish, we are done; if it does, factor out the Shalika local coefficients computed above.

Suppose π is a parahoric spherical regular generic representation of G admitting a Shalika model. By [16, Cor. 1.1] and Proposition 6, π is the normalized parabolic induction

$$\pi \cong \mathrm{Ind}_P^G\left(\prod_{i=1}^{a}(\theta_i\times\theta_i^{-1})\times\prod_{i=1}^{b}\varepsilon_i\,\mathrm{St}\times\prod_{i=1}^{c}(\eta_i\,\mathrm{St}\times\eta_i^{-1}\,\mathrm{St})\right),$$

where $\theta_i, \varepsilon_i, \eta_i$ are unramified characters, St is the Steinberg representation on GL_2, $\varepsilon_i^2 = 1$, and $a+b+2c = n$. By regularity $b \leqslant 2$, as the only possible ε_i that can appear are 1 and the unramified quadratic character.

Let θ be the unramified character of T such that $\theta_{a+i} = \varepsilon_i|\cdot|^{\frac{1}{2}}$ for $1 \leqslant i \leqslant b$, and $\theta_{a+b+2i-1} = \eta_i|\cdot|^{\frac{1}{2}}, \theta_{a+b+2i} = \eta_i^{-1}|\cdot|^{\frac{1}{2}}$ for $1 \leqslant i \leqslant c$; for $1 \leqslant i \leqslant n$, set $\theta_{n+i} = \theta_i^{-1}$. The assumption that π is regular translates to $\theta_i \neq \theta_j$ for all $i < j$. Then $\pi \subset \mathrm{Ind}_B^G \theta$ is the unique irreducible subrepresentation, by the Bernstein-Zelevinsky classification. We remark that the U-eigenvalue α is uniquely determined by θ, and each spin refinement $\widetilde{\pi}$ can be obtained in this way. Let $\tau \in W$ be given by the permutation

$$\tau = \prod_{i=1}^{b}(a+i, n+a+i)\prod_{i=1}^{c}(a+b+2i-1, n+a+b+2i-1)(a+b+2i, n+a+b+2i). \tag{7}$$

The intertwining operator $M_\tau : I(\theta^{\tau^{-1}}) \to I(\theta)$ has image π as, on each GL_2 Levi block, the intertwining $M_{(i,j)} : \mathrm{Ind}_{B_2}^{\mathrm{GL}_2}(\eta_i|\cdot|^{-\frac{1}{2}} \times \eta_i|\cdot|^{\frac{1}{2}}) \to \mathrm{Ind}_{B_2}^{\mathrm{GL}_2}(\eta_i|\cdot|^{\frac{1}{2}} \times \eta_i|\cdot|^{-\frac{1}{2}})$ has image $\eta_i \,\mathrm{St}$.

For a complex $2n$-tuple $s = (s_1, s_2, \ldots, s_{2n})$, consider $\theta_s = (|\cdot|^{s_1}\theta_1, \ldots, |\cdot|^{s_{2n}}\theta_{2n})$. For s in a Zariski open in $\mathbb{C}^{2n}$, the representation $I(\theta_s)$ is irreducible and $M_\tau : I(\theta_s^{\tau^{-1}}) \to I(\theta_s)$ is an isomorphism. Let $\mathcal{F}(\theta_s)$ be the vector space of linear combinations of flat families in $I(\theta_s)$, with coefficients which are holomorphic functions in s. Note that G acts on $\mathcal{F}(\theta_s)$ by admissibility, and $M_\tau : \mathcal{F}(\theta_s^{\tau^{-1}}) \to \mathcal{F}(\theta_s)$ is an isomorphism.

For $f \in \pi \subset I(\theta)$, consider $f_s \in \mathcal{F}(\theta_s)$, e.g., a flat family, with $f = f_0$. The naive definition $\mathcal{S}_{\mathrm{AG}}(f) = \lim_{s\to 0} \mathcal{S}_{\mathrm{AG}}^{\theta_s}(f_s)$ might not work, as it might happen that $\mathcal{S}_{\mathrm{AG}}(f)$ vanishes for all $f \in \pi$. In fact, this always happens if the monodromy rank of π is larger than 2. However, the computations in this section show that

$$\mathcal{S}_{\mathrm{AG}}^{\theta_s}(f_s) = \mathcal{C}_{\theta_s,\tau} \cdot \mathcal{S}_{\mathrm{AG}}^{\theta_s^{\tau^{-1}}}(M_\tau^{-1}(f_s)). \tag{8}$$

The function $\mathcal{S}_{\mathrm{AG}}^{\theta_s}(f_s)$ is holomorphic in s, but $\mathcal{S}_{\mathrm{AG}}^{\theta_s^{\tau^{-1}}}(M_\tau^{-1}(f_s))$ might not be. If it was, we could define

$$\mathcal{S}_\pi(f) = \lim_{s\to 0} \frac{\mathcal{S}_{\mathrm{AG}}^{\theta_s}(f_s)}{\mathcal{C}_{\theta_s,\tau}} = \lim_{s\to 0} \mathcal{S}_{\mathrm{AG}}^{\theta_s^{\tau^{-1}}}(M_\tau^{-1}(f_s)),$$

which we cannot show to be a Shalika functional. To account for this, we must take a preimage under the intertwining operator before considering families. We begin with a crucial technical result:

Proposition 17. *The kernel of the intertwining operator $M_\tau : I(\theta^{\tau^{-1}}) \to I(\theta)$ is contained in the kernel of the Ash–Ginzburg Shalika functional $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}} : I(\theta^{\tau^{-1}}) \to \mathbb{C}$.*

Proof. By the Bernstein-Zelevinsky classification, since $\mathrm{Im}\, M_\tau = \pi$, $(\ker M_\tau)^{\mathrm{ss}} = \bigoplus \sigma$, where σ runs over all components of $I(\theta)$ different from π. Our strategy is to realize $\ker M_\tau$ in the linear span of a collection of auxiliary intertwining operators.

Suppose $b \neq 0$ (otherwise this paragraph does not apply). For each $1 \leqslant i \leqslant b$, we will move the characters $\varepsilon_i|\cdot|^{\pm 1/2}$ to positions n and $2n$, swap, and move back. Let

$$\nu_i = (2n, 2n-1, \ldots, n+a+i)(n, n-1, \ldots, a+i)$$

and $\mu_i = \nu_i^{-1}(n, 2n)\nu_i$. The semisimplification of the image of $M_{\mu_i} : I(\theta^{\mu_i^{-1}\tau^{-1}}) \to I(\theta^{\tau^{-1}})$ will contain all components which are $\varepsilon_i \circ \det$ on the Levi block $(a+i, n+a+i)$. Indeed, since θ is regular, M_{ν_i} is an isomorphism, and the image of $M_{(n,2n)}$ will give the character on the Levi block.

By Proposition 10, we see that

$$\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}} \circ M_{\mu_i} = \mathcal{C}_{\theta^{\tau^{-1}},\mu_i} \mathcal{S}_{\mathrm{AG}}^{\theta^{\mu_i^{-1}\tau^{-1}}},$$

where $\mathcal{C}_{\theta^{\tau^{-1}},\mu_i}$ contains as factor

$$c_{n,2n}(\theta^{\nu_i^{-1}\tau^{-1}}) = c\left(\frac{\varepsilon_i|\cdot|^{-1/2}}{\varepsilon_i|\cdot|^{1/2}}\right) = 0.$$

We deduce that $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}$ vanishes on $\mathrm{Im}\, M_{\mu_i}$.

Suppose, now, that $1 \leqslant i \leqslant c$. In this case, we need two auxiliary intertwining operators. Consider the permutations $\nu_i = (2n, 2n-1, \ldots, n+a+b+2i)(n, n-1, \ldots, a+b+2i)$, $\mu_i = \nu_i^{-1}(n, 2n)\nu_i$, and $\mu_i' = \nu_i^{-1}(n, 2n)\nu_i(a+b+2i-1, a+b+2i)(n+a+b+2i-1, n+a+b+2i)$.

First, the semisimplification of the image of $M_{\mu_i} : I(\theta^{\mu_i^{-1}\tau^{-1}}) \to I(\theta^{\tau^{-1}})$ contains all components which are $\eta \circ \det$ on the Levi block $(a+b+2i, n+a+b+2i-1)$, as the only linked characters that swap order are $\eta|\cdot|^{\pm 1/2}$. Again, we see that $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}} \circ M_{\mu_i} = \mathcal{C}_{\theta^{\tau^{-1}},\mu_i} \mathcal{S}_{\mathrm{AG}}^{\theta^{\mu_i^{-1}\tau^{-1}}}$, where $\mathcal{C}_{\theta^{\tau^{-1}},\mu_i}$ contains as factor $c_{n,2n-1}(\theta^{\nu_i^{-1}\tau^{-1}}) = c\left(\frac{\eta_i|\cdot|^{-1/2}}{\eta_i|\cdot|^{1/2}}\right) = 0$, so $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}$ vanishes on $\operatorname{Im} M_{\mu_i}$.

Finally, the semisimplification of the image of $M_{\mu_i'} : I(\theta^{(\mu_i')^{-1}\tau^{-1}}) \to I(\theta^{\tau^{-1}})$ contains all components which are $\eta^{-1} \circ \det$ on the Levi block $(a+b+2i-1, n+a+b+2i)$, as the only linked characters that swap order are $\eta^{-1}|\cdot|^{\pm 1/2}$. Again, we see that $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}} \circ M_{\mu_i'} = \mathcal{C}_{\theta^{\tau^{-1}},\mu_i'} \mathcal{S}_{\mathrm{AG}}^{\theta^{(\mu_i')^{-1}\tau^{-1}}}$, where $\mathcal{C}_{\theta^{\tau^{-1}},\mu_i'}$ contains as factor $c_{n,2n-1}(\theta^{(a+b+2i-1,a+b+2i)(n+a+b+2i-1,n+a+b+2i)\nu_i^{-1}\tau^{-1}}) = c(\frac{\eta_i^{-1}|\cdot|^{-1/2}}{\eta_i^{-1}|\cdot|^{1/2}}) = 0$, so $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}$ vanishes on $\operatorname{Im} M_{\mu_i'}$.

We conclude that $\mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}$ vanishes on the linear span of all components of $I(\theta^{\tau^{-1}})$ other than π, and thus also on $\ker M_\tau$, as desired. □

Definition 18. For $f \in \pi$ let $g \in I(\theta^{\tau^{-1}})$ such that $f = M_\tau g$, and let g_s any analytic family such that $g_0 = g$. Define

$$\mathcal{S}_\pi(f) = \lim_{s\to 0} \mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}(g_s).$$

This will satisfy the requirements of Theorem 7. First, $\mathcal{S}_\pi(f)$ is well-defined. Indeed, if g' is another such preimage and g_s' is an analytic family through g' then

$$\lim_{s\to 0} \mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}(g_s) = \mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}(g) = \mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}(g') = \lim_{s\to 0} \mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}(g_s'),$$

as $g - g' \in \ker M_\tau \subset \ker \mathcal{S}_{\mathrm{AG}}^{\theta^{\tau^{-1}}}$ by Proposition 17. Second, $\mathcal{S}_\pi$ is a Shalika functional, being defined by the analytic continuation of Shalika functionals. Finally, if $f = M_\tau f_{Bw_0J}^{\theta^{\tau^{-1}}}$, then $g_s = f_{Bw_0J}^{\theta_s^{\tau^{-1}}}$ is a flat family, and

$$\mathcal{S}_\pi(f)(t^{-\delta}\varpi^\delta) = \lim_{s\to 0} \mathcal{S}_{\mathrm{AG}}^{\theta_s^{\tau^{-1}}}(f_{Bw_0J}^{\theta^{\tau^{-1}}})(t^{-\delta}\varpi^\delta) = 1.$$

3. Parahoric level p-adic L-functions and p-adic families

We first recall some automorphic results from [9], then we present the Friedberg–Jacquet style linear functional on the overconvergent coholmology constructed in [4] while setting up global notations.

Let F be a totally real number field of degree d, let $\mathcal{O}_F$ be its ring of integers and Σ the set of its real embeddings. Let $\mathbb{A} = \mathbb{A}_f \times \mathbb{R}$ denote the ring of adeles of $\mathbb{Q}$. For v a non-archimedean place of F, we let F_v be the completion of F at v, denote by $\mathcal{O}_v$ the ring of integers in F_v, and fix a uniformiser ϖ_v.

Let $n \geqslant 1$ and let G denote the algebraic group $\mathrm{Res}_{\mathcal{O}_F/\mathbb{Z}}\mathrm{GL}_{2n}$, $B = \mathrm{Res}_{\mathcal{O}_F/\mathbb{Z}}B_{2n}$ for the Borel subgroup of upper triangular matrices, N its the unipotent radical, and $T = \mathrm{Res}_{\mathcal{O}_F/\mathbb{Z}}T_{2n}$ be the maximal split torus of diagonal matrices. Let $H = \mathrm{Res}_{\mathcal{O}_F/\mathbb{Z}}(\mathrm{GL}_n \times \mathrm{GL}_n)$ which we diagonally embed into G. Consider the Siegel parabolic subgroup Q of G whose Levi subgroup is H, and let U be its unipotent radical.

Fix a rational prime p and an embedding $\iota_p : \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_p$. It determines partition of Σ into subsets $\Sigma_{\mathfrak{p}}$ indexed by the set Σ_p of primes $\mathfrak{p}$ of F above p.

3.1. **The automorphic setup.** Let π be a cuspidal automorphic representations of $\mathrm{GL}_{2n}(\mathbb{A}_F)$ having central character ω. We assume that π is regular algebraic, meaning that it is cohomological of weight $\lambda = (\lambda_\sigma)_{\sigma\in\Sigma}$ which is dominant, integral and pure *i.e.* there exists $\mathsf{w} \in \mathbb{Z}$, the purity weight, such that for all $\sigma \in \Sigma$, $\lambda_{\sigma,1} \geqslant \cdots \geqslant \lambda_{\sigma,2n}$ are in $\mathbb{Z}$ and $\lambda_{\sigma,i} + \lambda_{\sigma,2n-i+1} = \mathsf{w}$ for all $1 \leqslant i \leqslant 2n$.

We further assume that π is essentially self-dual of symplectic type with respect to a Hecke character η or, equivalently by Asgari–Shahidi [1], that it is a functorial transfer of a globally generic representation of GSpin_{2n+1} having central character η, where we recall that $\eta^n = \omega$ and that $\eta|\cdot|^{\mathsf{w}}$ has finite order. Of utmost importance to us will be a third equivalent characterization, namely that π admits a Shalika model with respect to η. We call such a π a RASCAR. We also recall that this implies that for each place v of F, π_v admits a Shalika model with respect to η_v.

In this paper, we will assume that π is parahoric spherical, *i.e.* $\pi_v^{J_v} \neq \{0\}$ for each finite place v of F, where the parahoric subgroup J_v of $\mathrm{GL}_{2n}(\mathcal{O}_v)$ consists of matrices whose reduction modulo ϖ_v belongs to

Q. This implies that η_v is unramified, hence there exists an unramified character ξ_v such that $\xi_v^2 = \eta_v^{-1}$ and the results from §1-§2 apply to $\pi_v \otimes \xi_v$ which admits a Shalika model with respect to the trivial character.

As recalled in §2.1, π_v contributes to the (unitarily normalized) parabolic induction $\mathrm{Ind}_{B(F_v)}^{G(F_v)}\theta_v$ of an unramified character θ_v and we let $\alpha_{v,i} = \theta_{v,i}(\varpi_v)$ denote the Satake parameters. As $\pi_v \simeq \pi_v^\vee \otimes \eta_v$, the Satake parameters can be paired up with product $\eta(\varpi_v)$ in each couple.

Let $S = \{v \nmid p\infty : \pi_v \text{ not spherical}\}$ be the set of bad places for π. Both p-adic deformations and constructions of p-adic L-functions require a preliminary step consisting in the choice of p-adic Q-refinement $\widetilde{\pi} = \big(\pi, (\alpha_v)_{v\in S\cup S_p}\big)$ of π, where α_v is a U_v-eigenvalue on $\pi_v^{J_v}$, and it is essential to require the refinement $\widetilde{\pi}_v = (\pi_v, \alpha_v)$ to be spin (see [3, §6]). By [9] one has $\alpha_v = q_v^{n^2/2}\prod_{i\in I}\alpha_{v,i}$, where $q_v = \mathrm{N}_{F/\mathbb{Q}}(v)$ and I is a subset of $\{1,\dots,2n\}$ of cardinality n, and being spin amounts to choosing in I exactly one Satake parameter per couple with product $\eta(\varpi_v)$.

We will always assume the refinement $\widetilde{\pi}_v = (\pi_v, \alpha_v)$ to be regular, *i.e.* that the generalized eigenspace $\pi_v^{J_v}[\![U_v - \alpha_v]\!]$ is a line. As the refinement is spin this already implies that $\alpha_{v,i} \neq \alpha_{v,j}$ for all $i \in I, j \notin I$, but for simplicity we will further assume θ_v to be regular, *i.e.* the $\alpha_{v,i}$ to be pairwise distinct.

The explicit Shalika functionals for unramified principal series constructed by Ash–Ginzburg [2] play a central role in our approach. They specifically require the induced character to be ordered so that:

$$\theta_{v,i}\theta_{v,n+i} = \eta \text{ for all } \ 1 \leqslant i \leqslant n.$$

Henceforth we will consider our parahoric spherical π_v as a subrepresentation of $\mathrm{Ind}_{B(F_v)}^{G(F_v)}\theta_v$ with $\alpha_{v,i}\cdot\alpha_{v,n+i} = \eta(\varpi_v)$ and $\alpha_v = q_v^{n^2/2}\prod_{n+1}^{2n}\alpha_{v,i}$.

3.2. **Automorphic p-adic L-functions.**

Let $\widetilde{\pi}^S = (\pi, (\alpha_v)_{v\in S\cup S_p})$ be a regularly Q-refined RACAR of weight λ as in §3.1. Assume further that π admits a (η,ψ)-Shalika model:

$$\mathcal{S}_\psi^\eta : \pi \hookrightarrow \mathrm{Ind}_{S(\mathbb{A})}^{G(\mathbb{A})}(\eta\otimes\psi), \qquad \varphi \mapsto \mathcal{S}_\psi^\eta(\varphi) : g \mapsto \int_{Z_G(\mathbb{A})\mathcal{S}(\mathbb{Q})\backslash\mathcal{S}(\mathbb{A})} \varphi(sg)\ (\eta\otimes\psi)^{-1}(s)ds. \tag{9}$$

Consider the open compact subgroup $K(\widetilde{\pi}^S) = \prod_v K_v$ of $G(\mathbb{A}_f)$ where $K_v = \mathrm{GL}_{2n}(\mathcal{O}_v)$ is the maximal hyperspecial subgroup for $v \notin S\cup S_p$ and $K_v = J_v$ is the standard Q-parahoric subgroup for $v \in S\cup S_p$. Let $\mathcal{H}^S$ denote the spherical Hecke algebra away from $S\cup S_p$ and let $\mathfrak{m}_\pi$ be its maximal ideal attached to π (see [4, Def. 2.1], where this commutative algebra is denoted by $\mathcal{H}'$). Consider the following maximal ideal of $\mathcal{H}^S[U_v - \alpha_v, v\in S\cup S_p]$:

$$\mathfrak{m}_{\widetilde{\pi}}^S = (\mathfrak{m}_\pi, U_v - \alpha_v, v \in S\cup S_p).$$

Given a character ϵ of $\{\pm1\}^\Sigma$, we will now endow the line $\pi_f^{K(\widetilde{\pi})}[U_v - \alpha_v, v\in S\cup S_p]$ with two rational structures allowing us to define Betti–Shalika periods measuring the ratio between the two.

The locally symmetric space $S_K = G(\mathbb{Q})\backslash G(\mathbb{A})/KK_\infty^\circ$ is a $t = d(2n-1)(n+1)$-dimensional real orbifold. By [4, Prop. 2.3], there is a Hecke-equivariant isomorphism

$$\pi_f^{K(\widetilde{\pi}^S)}[U_v - \alpha_v, v\in S\cup S_p] \xrightarrow{\sim} \mathrm{H}_c^t(S_{K(\widetilde{\pi}^S)}, \mathscr{V}_\lambda^\vee(\overline{\mathbb{Q}}_p))_{\mathfrak{m}_{\widetilde{\pi}}^S}^\epsilon. \tag{10}$$

depending on the choice of basis of the relative Lie algebra cohomology and ι_p. The right hand side is a line admitting a basis $\phi_{\widetilde{\pi}^S}^\epsilon$ defined over the field of rationality of $\widetilde{\pi}^S$ (that is, the field of rationality of π to which we adjoin the α_v's) and which can be further assumed to be p-integral.

We will now use the Shalika model $\mathcal{S}_{\psi_f}^{\eta_f}$ of π_f to fix an appropriate basis of the left hand side.

For $v \notin S\cup S_p\cup\Sigma$, Friedberg–Jacquet [10] showed the existence of a spherical vector $W_v^{\mathrm{FJ}} \in \mathcal{S}_{\psi_v}^{\eta_v}(\pi_v)$ such that $W_v^{\mathrm{FJ}}(t_v^{-\delta_v}) = 1$. Moreover they show that for all unramified quasi-characters $\chi_v : F_v^\times \to \mathbb{C}^\times$, we have

$$\zeta_v\left(s+\tfrac12, W_v^{\mathrm{FJ}}, \chi_v\right) = \left[q_v^s\chi_v(\varpi_v)\right]^{n\delta_v}\cdot L\left(\pi_v\otimes\chi_v, s+\tfrac12\right). \tag{11}$$

For $v \in S\cup S_p$, by Theorem 7 there exists $W_{\widetilde{\pi}_v} \in \mathcal{S}_{\psi_v}^{\eta_v}(\pi_v^{J_v})[U_v - \alpha_v]$ such that $W_{\widetilde{\pi}_v}(t_v^{-\delta_v}) = 1$. Moreover by Proposition 8 for all unramified quasi-characters $\chi_v : F_v^\times \to \mathbb{C}^\times$, we have

$$\zeta(s+\tfrac12, W_{\widetilde{\pi}_v}, \chi_v) = q_v^{\delta_v\left[(s+\frac12)n - \frac{n^2}{2}\right]}\left(1-\frac{1}{q_v}\right)^n \chi_v(\varpi_v)^{-\delta_v n} L(\theta'_{n+i,v}\chi_v, s+\tfrac12).$$

A straightforward consequence of this is the existence of an explicit Friedberg–Jacquet test vector

$$W_{\pi_v} = \sum_{\alpha_v} \kappa_{\alpha_v} W_{\widetilde{\pi}_v}.$$

for π_v parahoric spherical, where the sum is over all possible refinements $\widetilde{\pi}_v = (\pi_v, \alpha_v)$ of π_v (assumed all regular) and the rational constants κ_{α_v} can be explicitly computed using partial fractions decompositions.

Thus $\left(\mathcal{S}_{\psi_f}^{\eta_f}\right)^{-1}\left(\otimes_{v\notin S\cup S_p\cup\Sigma} W_v^{\mathrm{FJ}} \otimes_{v\in S} W_{\pi_v} \otimes_{v\in S_p} W_{\widetilde{\pi}_v}\right)$ is a basis of the left hand side of (10) and we define the Betti–Shalika period $\Omega_{\widetilde{\pi}}^{\epsilon}$ as its coordinate in the basis $\phi_{\widetilde{\pi}}^{\epsilon} = \sum_{\alpha_S} \kappa_{\alpha_S} \phi_{\widetilde{\pi}^S}^{\epsilon}$ that we have chosen earlier.

Assuming that $\widetilde{\pi}$ is non-Q-critical in the sense of [4, Def. 3.14], *i.e.*, that the natural map on generalized eigenspaces

$$\mathrm{H}_c^\bullet(S_{K(\widetilde{\pi}^S)}, \mathscr{D}_\lambda(L))_{\mathfrak{m}_{\widetilde{\pi}}^S} \xrightarrow{\sim} \mathrm{H}_c^\bullet(S_{K(\widetilde{\pi}^S)}, \mathscr{V}_\lambda^\vee(L))_{\mathfrak{m}_{\widetilde{\pi}}^S}$$

is an isomorphism, allows us for each $\epsilon \in \{\pm 1\}^\Sigma$ to uniquely lift $\phi_{\widetilde{\pi}^S}^{\epsilon}$ to an U_p-eigenclass $\Phi_{\widetilde{\pi}^S}^{\epsilon} \in \mathrm{H}_c^t(S_{K(\widetilde{\pi}^S)}, \mathscr{D}_\lambda)_{\mathfrak{m}_{\widetilde{\pi}}}^{\epsilon}$. Recall that the U_p-eigenvalue on $\Phi_{\widetilde{\pi}^S}^{\epsilon}$ equals $\alpha_p^\circ = \lambda(t_p)\alpha_p$. Let $\Phi_{\widetilde{\pi}}^{\epsilon} = \sum_{\alpha_S} \kappa_{\alpha_S} \Phi_{\widetilde{\pi}^S}^{\epsilon}$.

Let $\mathcal{L}_p(\tilde{\pi}) = A^{-1}\cdot\mu^{\eta_0}(\Phi_{\tilde{\pi}})$ be the L-valued distribution on Gal_p attached to $\Phi_{\tilde{\pi}} = \sum_{\epsilon\in\{\pm1\}^\Sigma} \Phi_{\tilde{\pi}}^{\epsilon}$ constructed in [4, Def. 6.17], where A is the global constant from [4, Def. 6.23].

Theorem 19. *The distribution $\mathcal{L}_p(\tilde{\pi})$ is admissible of growth $h_p = v_p(\alpha_p^\circ)$. For every finite order Hecke character χ of F of conductor p^β, and all $j \in \mathrm{Crit}(\lambda)$, we have*

$$\iota_p^{-1}(\mathcal{L}_p(\tilde{\pi}, \chi\chi_{\mathrm{cyc}}^j)) = \mathcal{G}(\chi_f)^n \mathrm{N}_{F/\mathbb{Q}}(-i\mathfrak{d})^{jn} \prod_{\mathfrak{p}\in\Sigma_p} e_{\mathfrak{p}}(\tilde{\pi}_{\mathfrak{p}}, \chi_{\mathfrak{p}}, j) \cdot \frac{L^{(p)}\left(\pi\otimes\chi, j+\frac{1}{2}\right)}{\Omega_{\tilde{\pi}}^{\epsilon}}, \tag{12}$$

where $\epsilon = (\chi\chi_{\mathrm{cyc}}^j\eta)_\infty$, $\mathcal{G}(\chi_f)$ is the Gauss sum and

$$e_{\mathfrak{p}}(\tilde{\pi}_{\mathfrak{p}}, \chi_{\mathfrak{p}}, j) = \begin{cases} \left(q_{\mathfrak{p}}^{nj+\binom{n}{2}}\alpha_{\mathfrak{p}}^{-1}\right)^{\beta_{\mathfrak{p}}} & : \chi_{\mathfrak{p}} \text{ ramified,} \\ \prod_{i=n+1}^{2n} \frac{1-(\theta'_{\mathfrak{p},i}\chi_{\mathfrak{p}})^{-1}(\varpi_{\mathfrak{p}})q_{\mathfrak{p}}^{j-1/2}}{1-\theta'_{\mathfrak{p},i}\chi_{\mathfrak{p}}(\varpi_{\mathfrak{p}})q_{\mathfrak{p}}^{-j-1/2}}. & : \chi_{\mathfrak{p}} \text{ unramified.} \end{cases}$$

3.3. **The eigenvariety and families of p-adic L-functions.** Here we follow closely [4, §8.3]. Let π be a RASCAR of weight λ which is parahoric spherical at every finite place and recall that S denotes the set of bad places outside p. Let $\widetilde{\pi}^S$ be a (regular) non-Q-critically refinement as in §3.2. Let Ω be a sufficiently small affinoid neigbourhood of λ in the parahoric weight space $\mathcal{W}^Q$ such that the overconvergent cohomology $\mathrm{H}_c^t(S_{(\widetilde{\pi}^S)}, \mathcal{D}_\Omega)$ admit slope decomposition with respect to the compact operator U_p. It is endowed with an $\mathcal{O}(\Omega)$-linear action of the Hecke algebra $\mathcal{H}^S[U_v - \alpha_v, v \in S\cup S_p]$ yielding for each $h \in \mathbb{Q}_{\geqslant 0}$ a local piece $\mathcal{E}_{\Omega,h}^S$ of an eigenvariety $\mathcal{E}^S$. While this eigenvariety slightly differs from the one considered in *loc. cit.* in that, locally at $v \in S$ we have put the Hecke operator U_v instead of diamond like operators S_v, all the constructions and arguments transfer *mutatis mutandis* yielding the following theorem. Indeed, the cyclicity follows from the fact that we use top degree cohomology, while the étaleness follows from the local multiplicity for the family ensured by (10) being a line. The proof itself is a subtle interplay between the closeness of the condition of being Spin and the openness of the condition of being Shalika – a major theme in *loc. cit.* to which we refer to more details.

Theorem 20. *Suppose that λ is regular and that $\tilde{\pi}^S$ is a regular, strongly non-Q-critical refinement in the sense of* [4, Def. 3.14]. *Then, after possibly shrinking Ω, the weight map $\mathsf{w} : \mathcal{E}^S \to \mathcal{W}^Q$ is étale at the point $x_{\tilde{\pi}^S}$ attached to $\tilde{\pi}^S$, i.e., there is a connected component $\mathcal{C}$ of $\mathcal{E}^S$ through $x_{\tilde{\pi}^S}$ such that $\mathsf{w} : \mathcal{C} \xrightarrow{\sim} \Omega$.*

Moreover, for each $\epsilon \in \{\pm1\}^\Sigma$, there exists a Hecke eigenclass $\Phi_{\mathcal{C}}^{\epsilon} \in \mathrm{H}_c^t(S_{K(\widetilde{\pi}^S)}, \mathcal{D}_\Omega)^{\epsilon}$ such that for a Zariski dense set $\mathcal{C}^{\mathrm{cl}}$ of $x \in \mathcal{C}$ corresponding to non-Q-critical refined parahoric spherical RASCARs $\tilde{\pi}_x^S$, the specialization of $\Phi_{\mathcal{C}}^{\epsilon}$ at $\mathsf{w}(x)$ generates $\mathrm{H}_c^t(S_{K(\widetilde{\pi}^S)}, \mathcal{D}_{\mathsf{w}(x)})^{\epsilon}$.

The above theorem implies existance of p-adic periods $c_x^{\epsilon} \in L^\times$ such that the specialization of $\Phi_{\mathcal{C}}^{\epsilon}$ at $\mathsf{w}(x)$ equals $c_x^{\epsilon}\cdot\Phi_{\tilde{\pi}_x^S}^{\epsilon}$. It also allows to define a $\mathcal{O}(\Omega)$-valued distribution $\mathcal{L}_p^{\mathcal{C},\epsilon} = A^{-1}\cdot\mu^{\eta_0}(\Phi_{\mathcal{C}}^{\epsilon})$ on Gal_p via [4, Def. 6.17], whose specialization at any $x \in \mathcal{C}^{\mathrm{cl}}$ equals $c_x^{\epsilon}\cdot\mathcal{L}_p^{\epsilon}(\tilde{\pi}_x^S)$. The crucial point to check is the existence of Ω such that $\tilde{\pi}_x^S$ is regular for all $x \in \mathcal{C}^{\mathrm{cl}}$ so that Theorem 19 requiring regularity of the

induced character at all $v \in S \cup S_p$ applies. This is ensured by the regularity of π_v^S via the following Galois theoretic argument. There exists a $\mathcal{O}_\Omega$-valued pseudo-representation of Gal_F (or more precisely a Chenevier determinant) interpolating the Galois representations ρ_x attached to $x \in \mathcal{C}^{\mathrm{cl}}$ (when ρ_π is irreducible there is even a $\mathcal{O}_\Omega$-valued representation). By the Local-Global Compatibility, which is known in our case, the Satake parameters of the inducing character of $\pi_{x,v}$ can be read off the Weil–Deligne representation attached to the restriction of ρ_x at v. The continuity of the pseudo-representation and the regularity of π_v then ensures the claim.

In closing, let us observe that while the p-adic L-function $\mathcal{L}_p^\epsilon(\tilde{\pi}^S)$ is not uniquely determined by its interpolation property for $\tilde{\pi}^S$ of critical slope (but still non-Q-critical), the p-adic L-function for the family $\mathcal{L}_p^{\mathcal{C},\epsilon}$ is always uniquely determined by interpolation because of the density of the non-critical slope classical points.

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